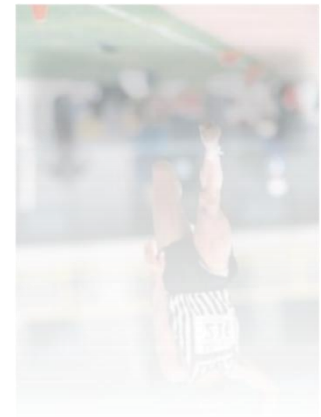


2.1 Linear Motion

2.2 Uniformly accelerated motion



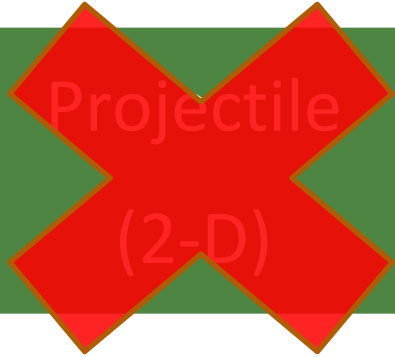
OVERVIEW: Kinematics of Linear Motion
studies of motion of an objects without
considering the effects that produce the
motion $\neq$ **DYNAMICS**

Linear/ Straight line
(1-D)

Horizontal with constant
acceleration
(x-axis only)

Vertical with constant
acceleration - Free Fall
(y-axis only)

Projectile
(2-D)





2.1 LINEAR MOTION

At the end of this chapter, students should be able to:

- (a) Define**
 - i) instantaneous velocity, average velocity and uniform velocity**
 - ii) instantaneous acceleration, average acceleration and uniform acceleration**
- (b) Compare the following quantities**
 - i) instantaneous velocity, average velocity and uniform velocity**
 - ii) instantaneous acceleration, average acceleration and uniform acceleration**
- (c) Sketch graphs of displacement-time, velocity-time and acceleration-time.**
- (d) Interpret graphs of displacement-time, velocity-time and acceleration-time.**
- (e) Determine the distance travelled, displacement, velocity and acceleration from appropriate graphs.**



2.2 UNIFORMLY ACCELERATED MOTION

At the end of this chapter, students should be able to:

- **Apply equations of motion with uniform acceleration:**

$$\begin{aligned}v &= u + at \\s &= ut + \frac{1}{2}at^2 \\v^2 &= u^2 + 2as\end{aligned}$$

- **Apply equations of motion for free fall:**

$$\begin{aligned}v &= u - gt \\s &= ut - \frac{1}{2}gt^2 \\v^2 &= u^2 - 2gs\end{aligned}$$

2.1 Linear Motion

Linear motion – motion of an object along a straight line path.

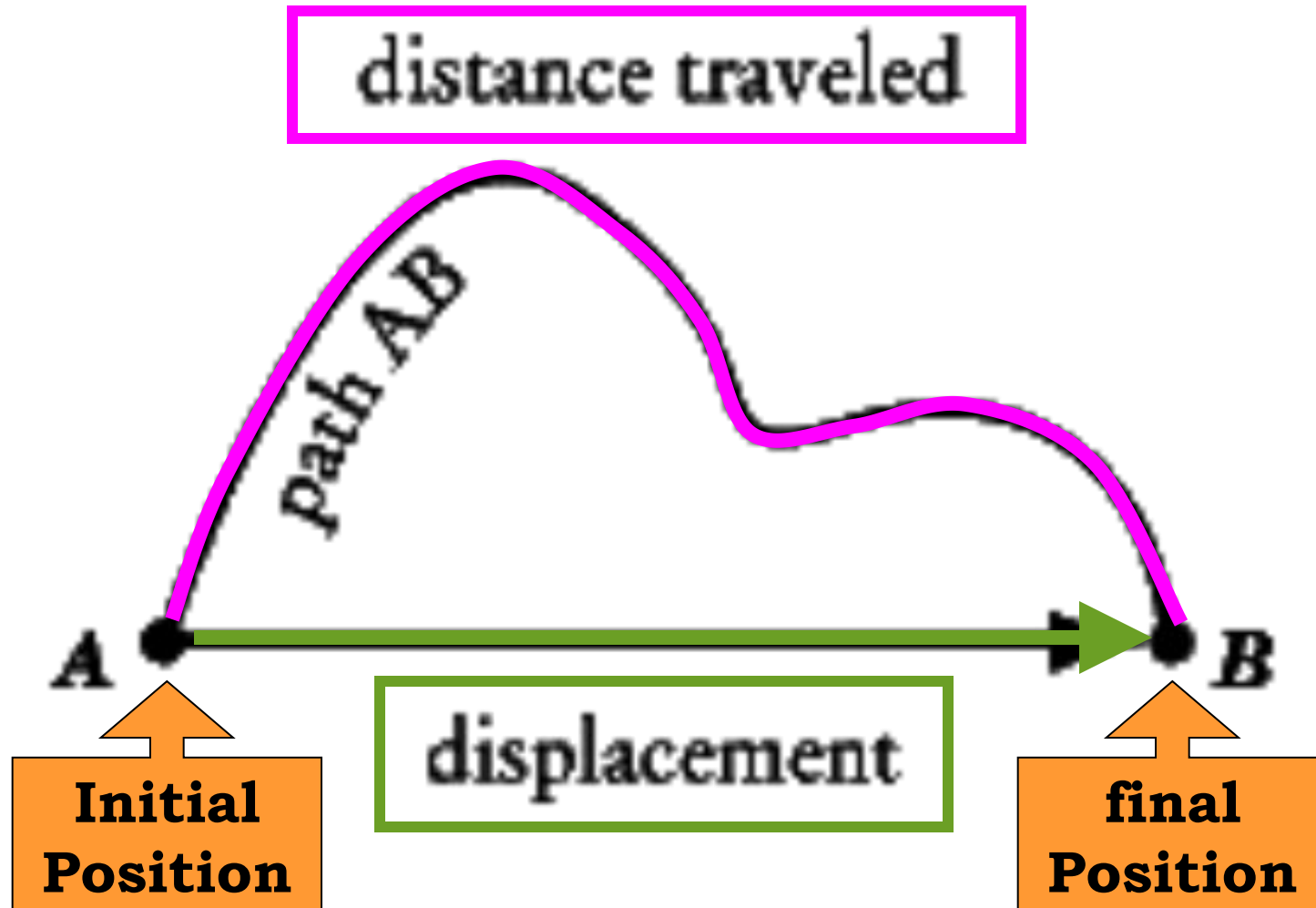
Distance, d

- ❑ defined as the **total length of travel** in moving from one location to another.
- ❑ **scalar** quantity.
- ❑ always positive.

Displacement, s

- ❑ **straight line distance** (shortest distance) from the initial position to the final position of an object.
- ❑ **Vector** quantity
- ❑ can be positive, negative @ zero.





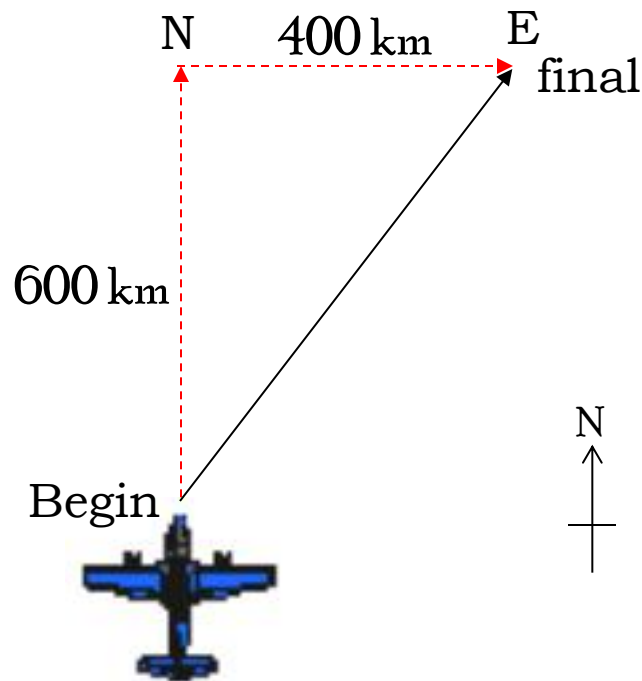
Example 1

Distance travelled = 200m

**Displacement = 120 m, in the direction of
Northeast**

Example 2

An air plane flies 600 km north and then 400 km to the east. What is the distance and displacement travelled by this air plane?



Total distance traveled $d = 600 + 400 = 1000$ km

Displacement, $s = \sqrt{600^2 + 400^2} = 721.11$ km

Speed, v

- defined as **change of distance per unit time** or **the rate of change of distance**.

$$v = \frac{\text{distance traveled}}{\text{time to travel that distance}} = \frac{d}{t}$$

- **scalar** quantity ; S.I. unit : **m s⁻¹**.

Average Speed, ($\bar{v}$)

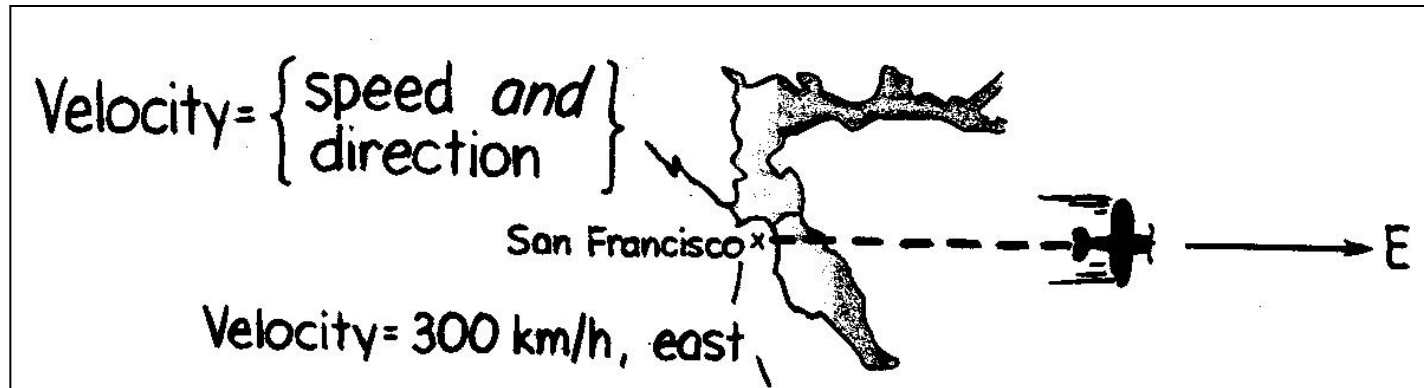
- defined as the **distance travelled divided by the elapsed time**.

$$\text{Average speed, } \bar{v} = \frac{\Sigma d}{\Sigma t}$$

- **Scalar** quantity ; S.I. unit : **m s⁻¹**

Velocity, v

- ❑ tells us how fast object is moving & in which direction it is moving.



- ❑ defined as **change of displacement per unit time** or **the rate of change of displacement**

$$\text{velocity, } \vec{v} = \frac{\text{displacement}}{\text{travel time}} = \frac{s}{t}$$

- ❑ **vector** quantity ; **SI unit : m s^{-1}**

Average velocity, v_{av}

$$\vec{v}_{av} = \frac{\text{change in displacement}}{\text{time taken for the change}}$$

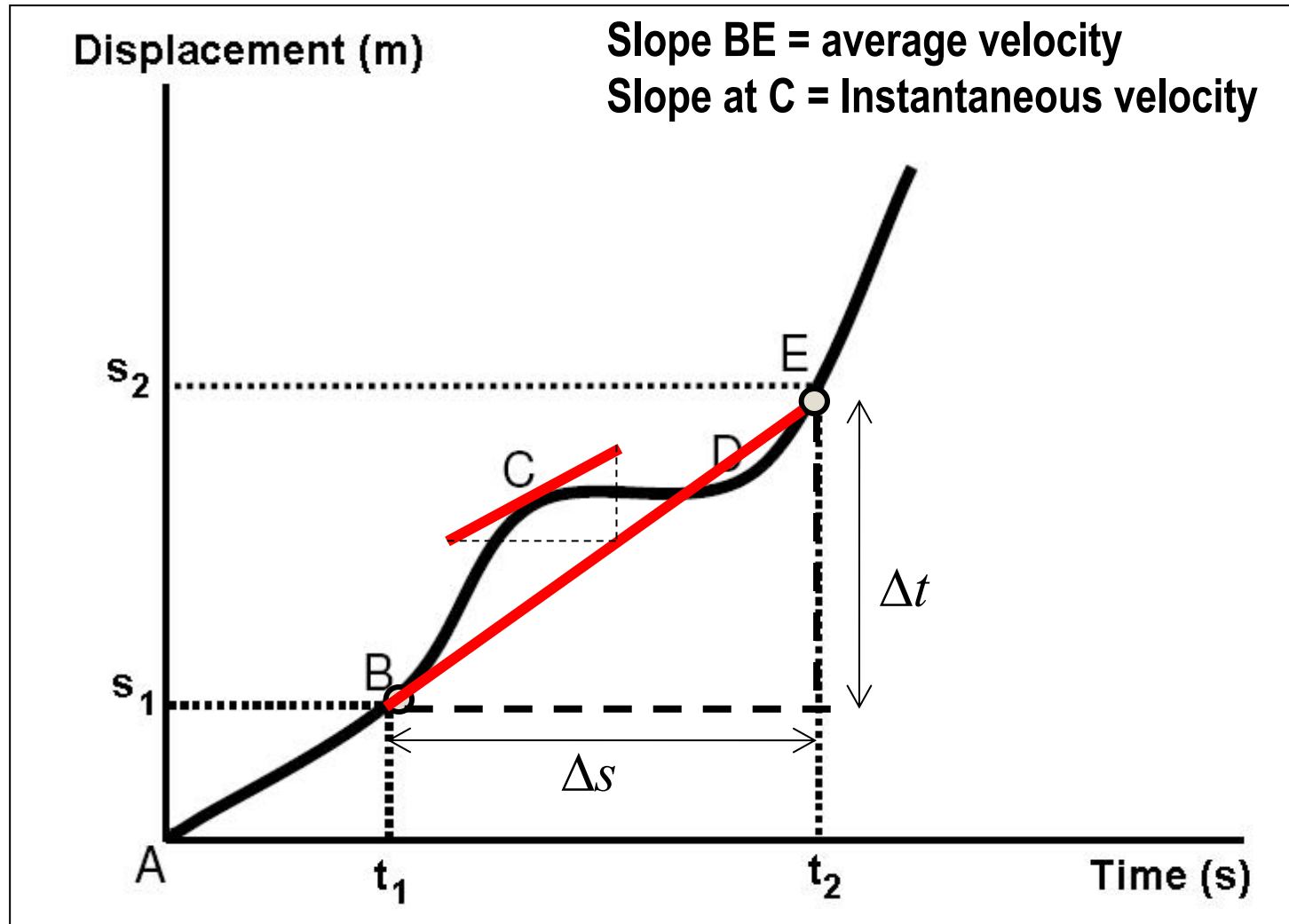
$$\vec{v}_{av} = \frac{\Delta s}{\Delta t} = \frac{s_2 - s_1}{t_2 - t_1}$$

Instantaneous velocity, v

- ❑ velocity **at a specified position** or **instant of time** along the path of motion.
- ❑ commonly referred as 'velocity at point A' or 'velocity at time t'.

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$$

- equal to the **gradient** at any point on the curve of a displacement – time ($s - t$) graph.



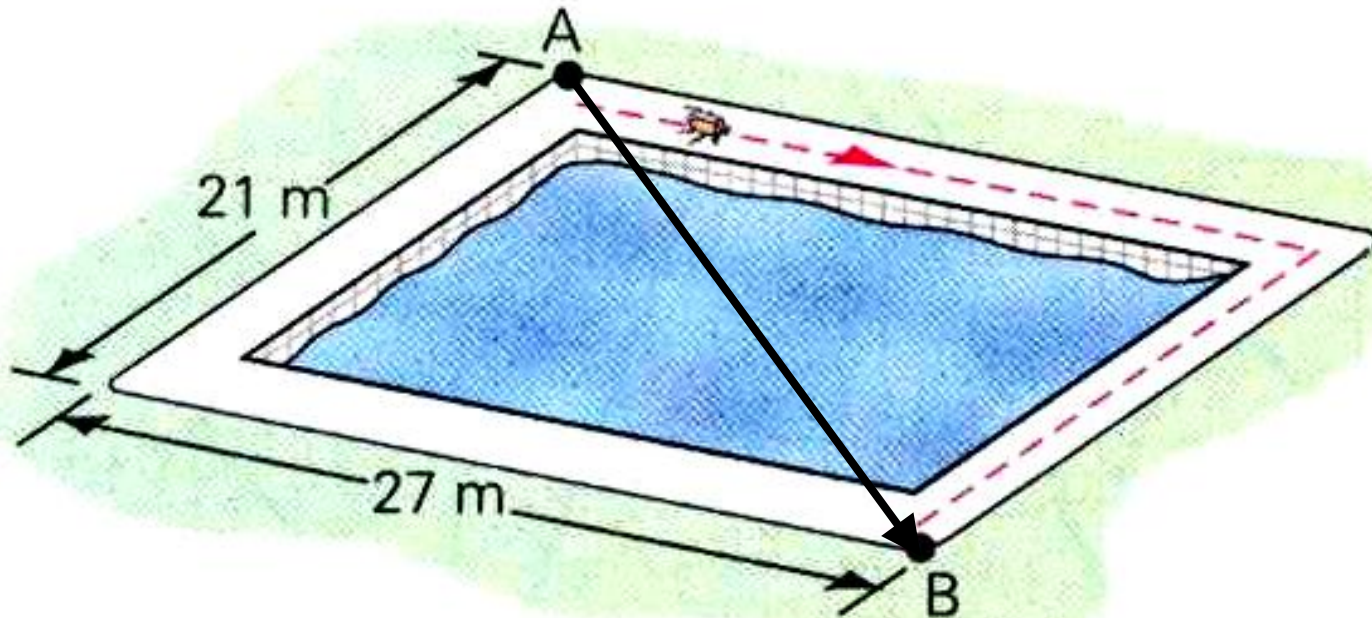
Uniform Velocity

- ❑ Velocity is uniform (**constant**) if its
 - a) magnitude is constant.
 - b) direction remain unchanged.
- ❑ The average and instantaneous velocities are equal.

Example 3

An insect crawls along the edge of a rectangular swimming pool of length 27m & width 21m. If it crawls from corner A to corner B in 30 min.

- (a) What is its average speed ?
- (b) What is the magnitude of its average velocity ?



Solution

Given : length $L = 27\text{m}$; width, $W = 21\text{m}$

Total distance travel, $d = 27 + 21 = 48\text{ m}$

Displacement, $s = \sqrt{27^2 + 21^2} = 34.21\text{ m}$

(a) *Average speed, $\bar{v} = \frac{\Sigma d}{\Sigma t}$*

$$= \frac{48}{30(60)}$$
$$= 0.0267\text{ m s}^{-1}$$



(b) *Average velocity*, $\vec{v}_{av} = \frac{\Delta s}{\Delta t}$



$$= \frac{34.21}{30(60)}$$
$$= 0.019 \, m \, s^{-1}$$

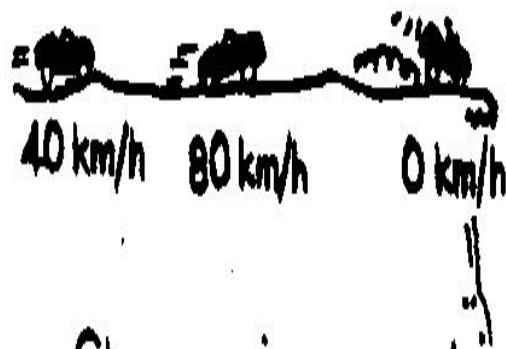
Acceleration, a

- defined as the **rate of change of velocity**

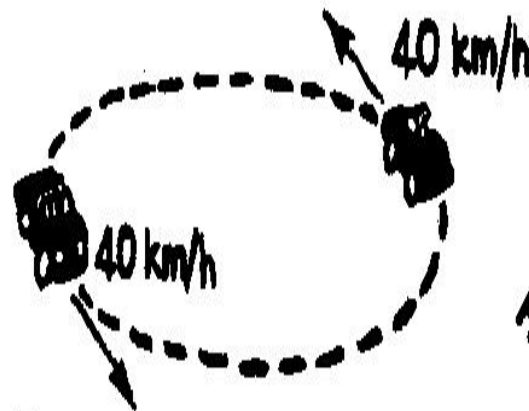
$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

- Velocity is **vector quantity**, $\rightarrow$ a change in velocity may thus involve either or both magnitude & direction.
- An acceleration may due to:
 - 1) **change in speed** (magnitude),
 - 2) **change in direction** or
 - 3) **change in both speed and direction**.
- **SI unit : m s^{-2}**

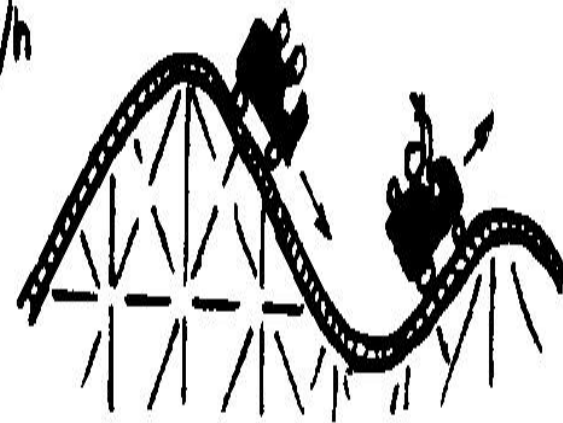
Acceleration = $\left\{ \begin{array}{l} \text{Rate of} \\ \text{change in} \\ \text{velocity} \end{array} \right\}$ due to $\left\{ \begin{array}{l} \text{change in speed} \\ \text{and/or direction} \end{array} \right\}$



Change in speed
but *not* direction



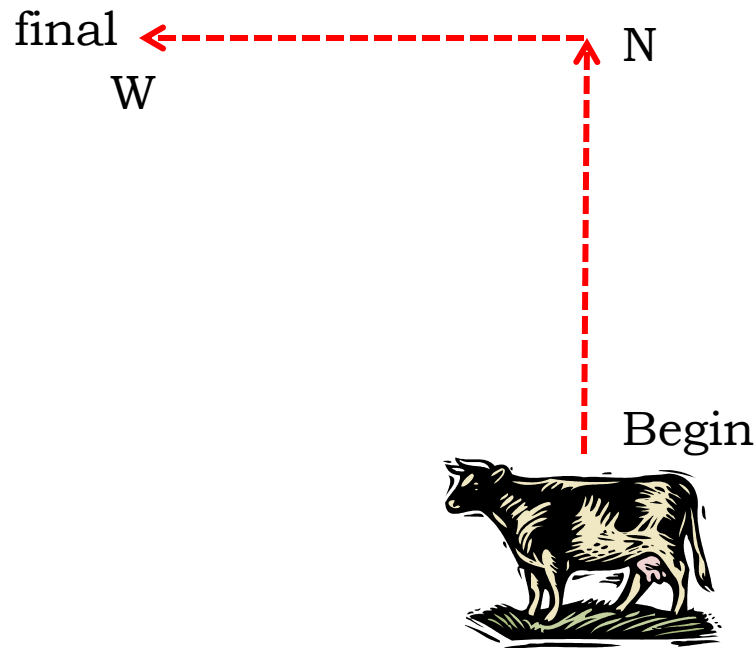
Change in direction
but *not* speed



Change in speed
and direction

Example 4

A cow is moving at 30 km h^{-1} to the north. Then it turns to the west without changing its speed. Is the cow accelerating?



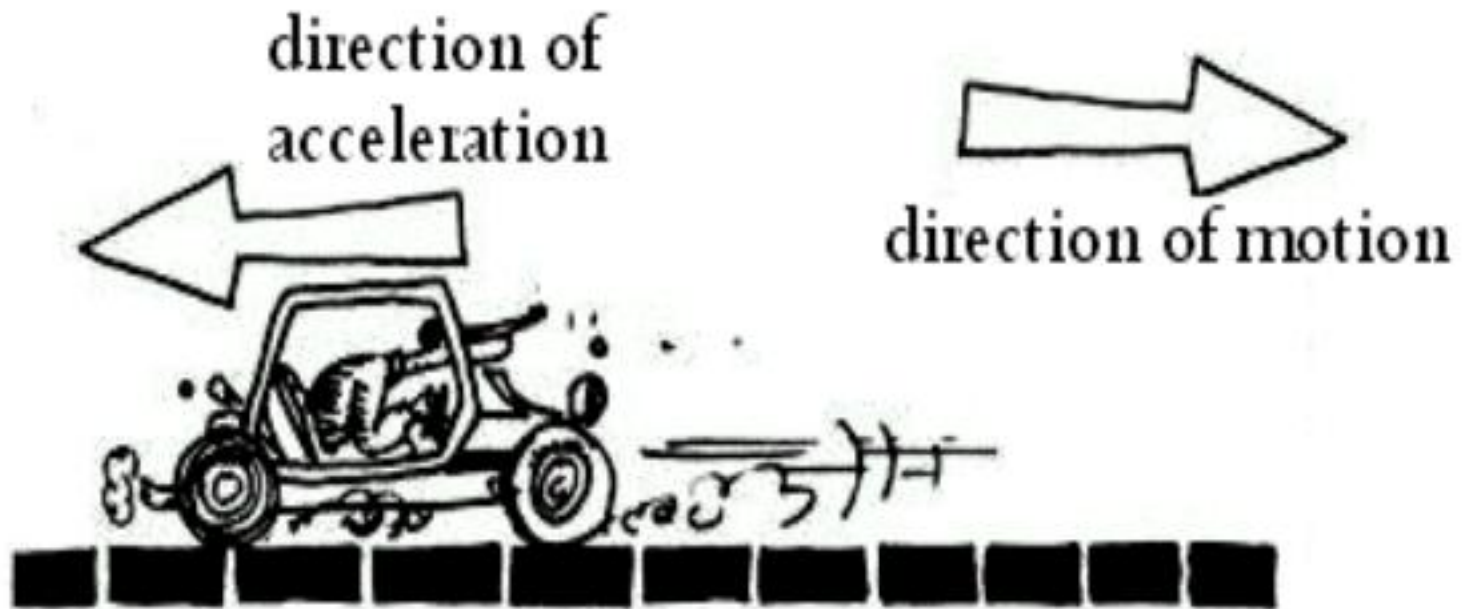
Answer : YES !

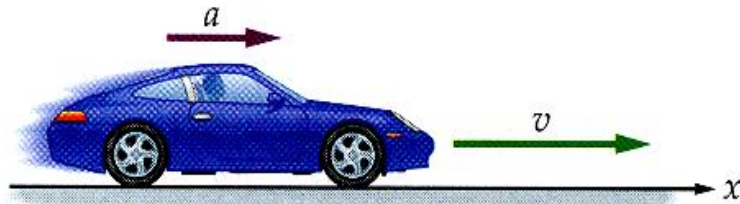
Reason : there is a change in direction

Deceleration :

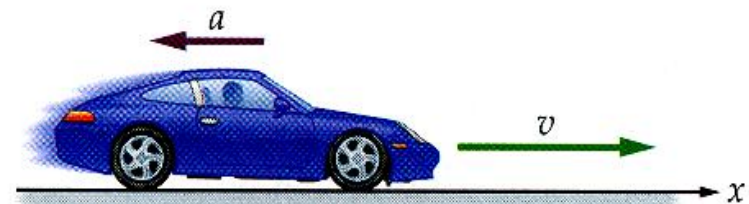
object is **slowing down**

(direction of acceleration is opposite to the direction of the motion or velocity).



Example 5

(a)



(b)



(c)



(d)



Acceleration $\rightarrow$ Car in figure (a) & (d)

Deceleration $\rightarrow$ Car in figure (b) & (c)

Average acceleration

- change in velocity divided by the time taken to make the change

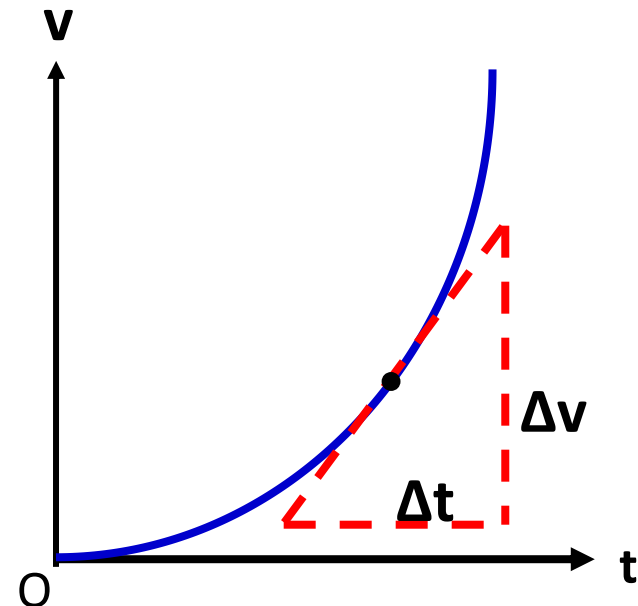
$$\vec{a} = \frac{\text{change in velocity}}{\text{time to make the change}} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta v}{\Delta t}$$

Instantaneous Acceleration

- acceleration at a particular instant of time.

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = \frac{d^2 s}{dt^2}$$

= the gradient
of the tangent



Uniform Acceleration

- ❑ Acceleration is uniform (**constant**) if its
 - a) magnitude is constant.
 - b) direction remain unchanged.
- ❑ The average and instantaneous accelerations are equal.

Example 6

A car travels in a straight line along a road. Its distance, s is given as a function of time t by the equation:

$$s(t) = 2.4t^2 - 0.12t^3$$

- (a) Calculate the average velocity of the car for the time interval $t = 0$ s and $t = 10$ s.**
- (b) Calculate the instantaneous velocity of the car at $t = 5$ s.**
- (c) Calculate the instantaneous acceleration of the car at $t = 5$ s.**

Solution

Given : $s(t) = 2.4t^2 - 0.12t^3$

(a) At $t_1 = 0$ s, $s_1 = 0$ m

At $t_2 = 10$ s,

$$s_2 = 2.4(10)^2 - 0.12(10)^3 = 120 \text{ m}$$

Average velocity, $\vec{v}_{av} = \frac{s_2 - s_1}{t_2 - t_1} = \frac{120 - 0}{10 - 0}$

$$\vec{v}_{av} = 12 \text{ m s}^{-1}$$

(b) Instantaneous velocity, $v = \frac{ds}{dt}$

$$= \frac{d}{dt}(2.4t^2 - 0.12t^3)$$

$$v = 4.8t - 0.36t^2$$

$$\text{At } t = 5\text{s}, v = 4.8(5) - 0.36(5)^2$$

$$v = 15 \text{ m s}^{-1}$$

(c) Instantaneous acceleration,

$$a = \frac{dv}{dt} = \frac{d}{dt}(4.8t - 0.36t^2)$$

$$a = 4.8 - 0.72t$$

$$\text{At } t = 5\text{s}, a = 4.8 - 0.72(5)$$

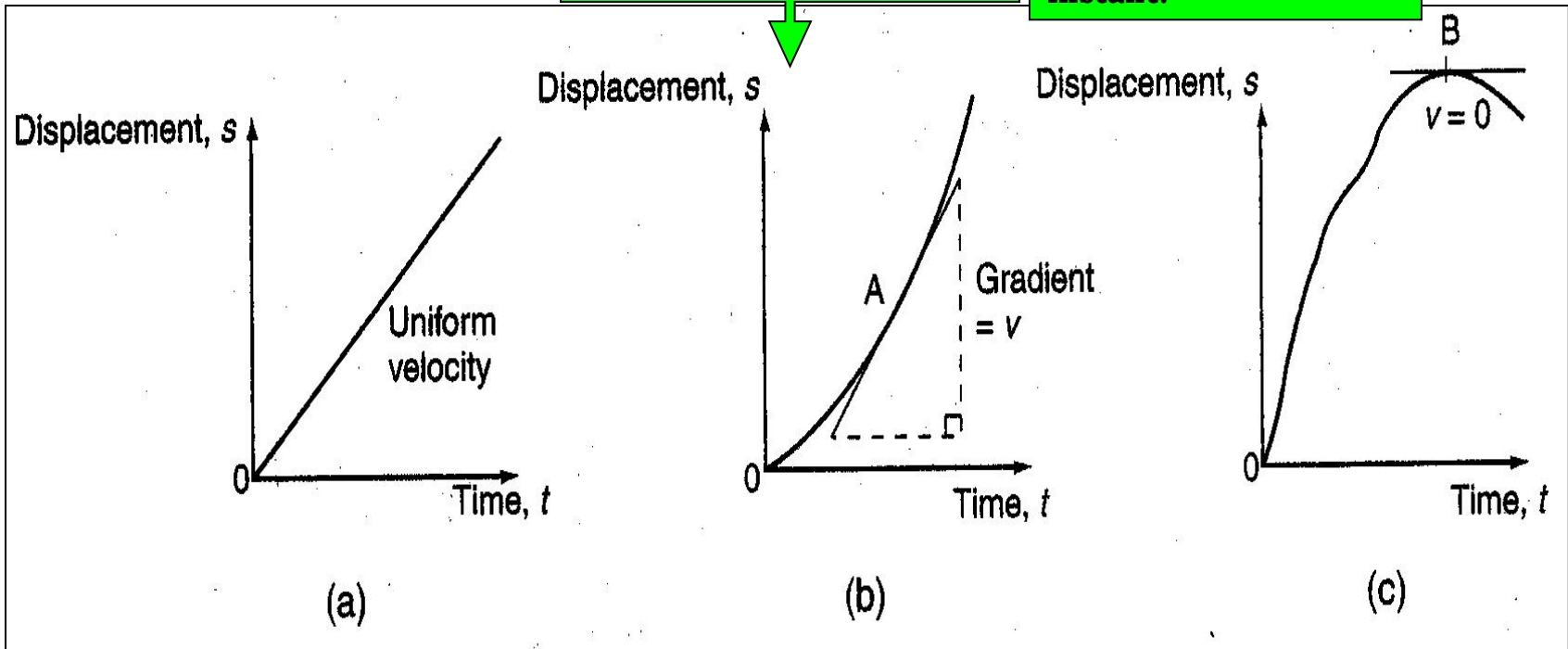
$$a = 1.2 \text{ m s}^{-2}$$

Graphical Methods

Displacement – time (s-t) graphs

We can determine velocity at point A by drawing a tangent line at A

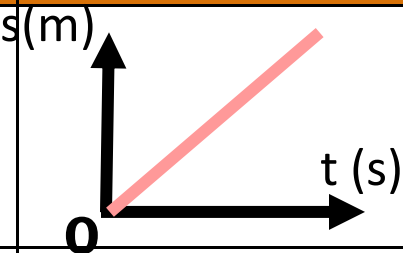
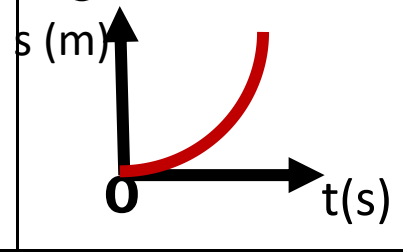
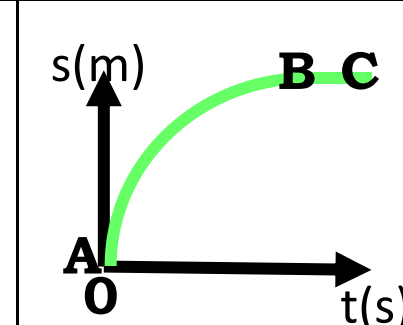
Gradient at B is zero which indicates that velocity = 0 at that instant.



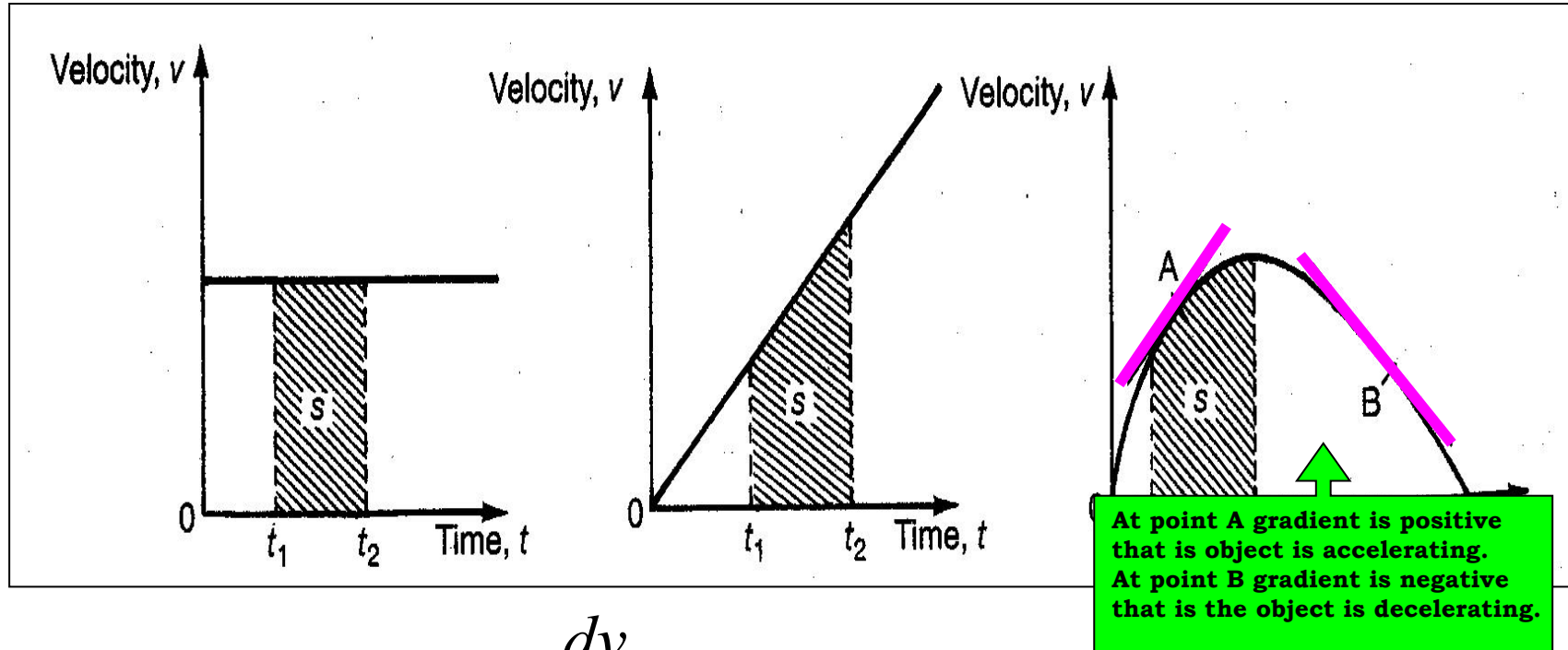
Instantaneous **velocity**, $v = \frac{ds}{dt}$ = **gradient** of (s-t) graph

Graphs for linear motion (s – t)

$$v = \frac{s_2 - s_1}{t_2 - t_1} = \frac{\Delta s}{\Delta t}$$

No	Type of Motion	Graph	Description
1	Uniform Motion		Velocity is constant.
2	Non Uniform Motion		The gradient increases, velocity increases $\Rightarrow$ Object is under acceleration.
3	Non Uniform Motion		- From point A to C the gradient decreases, velocity decreases $\Rightarrow$ Object is under deceleration. - Gradient at B to C is zero, velocity = 0

velocity – time (v-t) graphs



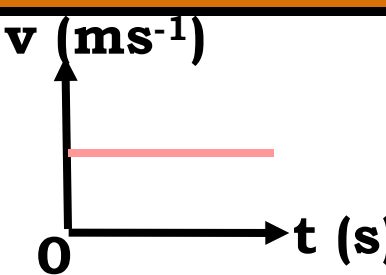
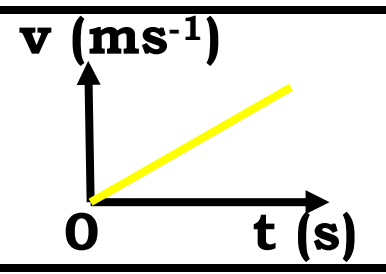
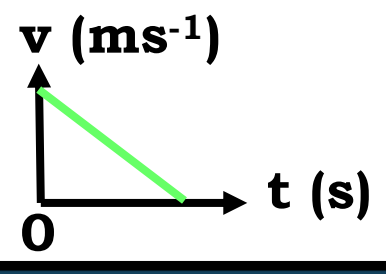
acceleration $a = \frac{dv}{dt}$ = **gradient** of (v-t) graph

Displacement of the object = shaded **area**
under the (v-t) graph

Graphs for linear motion (v – t)

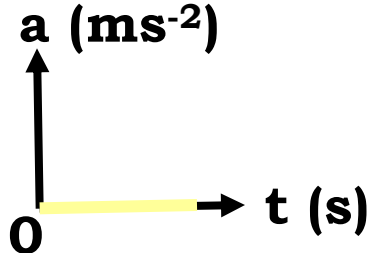
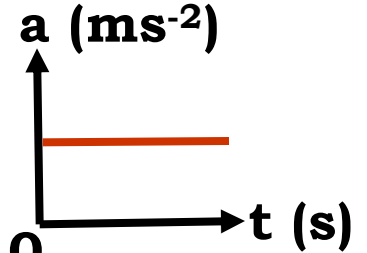
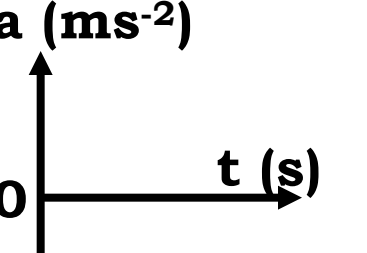
❖ Area under graph=Displacement

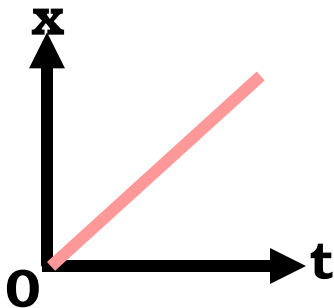
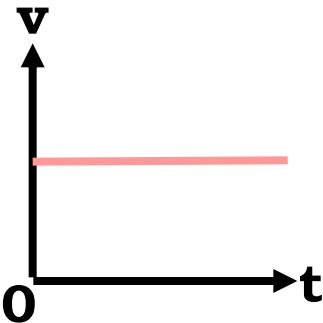
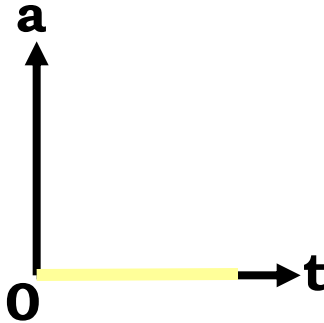
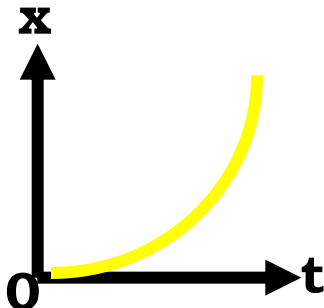
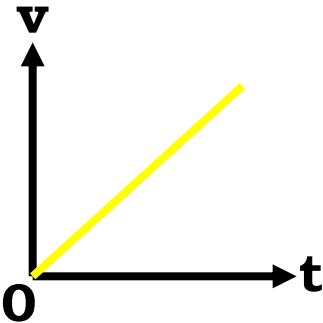
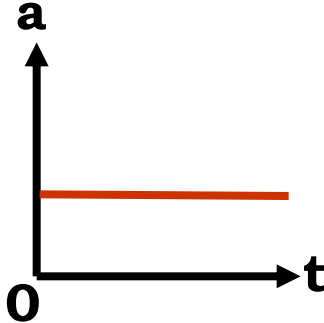
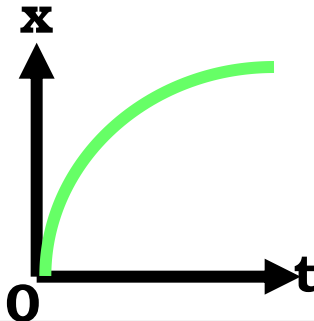
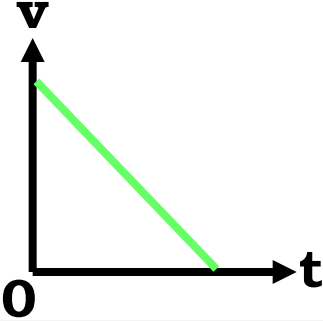
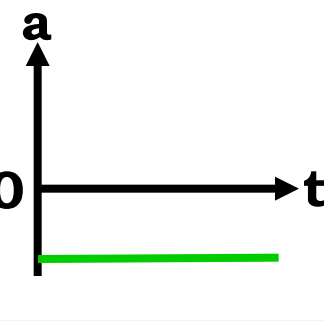
$$a = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta v}{\Delta t}$$

No	Type of Motion	Graph	Description
1	Uniform Motion	 A velocity-time graph with velocity v in ms^{-1} on the vertical axis and time t in s on the horizontal axis. The origin is marked 0. A horizontal red line is drawn at a constant positive velocity value.	Gradient = Acceleration=0
2	Non Uniform Motion	 A velocity-time graph with velocity v in ms^{-1} on the vertical axis and time t in s on the horizontal axis. The origin is marked 0. A straight yellow line starts from the origin and extends upwards with a constant positive gradient.	Velocity increases uniformly $\Rightarrow$ Constant acceleration
3	Non Uniform Motion	 A velocity-time graph with velocity v in ms^{-1} on the vertical axis and time t in s on the horizontal axis. The origin is marked 0. A straight green line starts from a positive value on the vertical axis and extends downwards to the horizontal axis with a constant negative gradient.	Velocity decreases uniformly $\Rightarrow$ Constant deceleration

Graphs for linear motion (a – t)

Area under graph = velocity

No	Type of Motion	Graph	Description
1	Uniform Motion	 The graph shows acceleration a (in ms^{-2}) on the vertical axis and time t (in s) on the horizontal axis. The vertical axis is labeled a (ms^{-2}) and the horizontal axis is labeled t (s). The origin is marked with 0. A horizontal yellow line is drawn along the t-axis, representing zero acceleration.	Acceleration is 0
2	Non Uniform Motion	 The graph shows acceleration a (in ms^{-2}) on the vertical axis and time t (in s) on the horizontal axis. The vertical axis is labeled a (ms^{-2}) and the horizontal axis is labeled t (s). The origin is marked with 0. A horizontal red line is drawn at a positive value on the a-axis, representing constant positive acceleration.	Object is under acceleration (positive value).
3	Non Uniform Motion	 The graph shows acceleration a (in ms^{-2}) on the vertical axis and time t (in s) on the horizontal axis. The vertical axis is labeled a (ms^{-2}) and the horizontal axis is labeled t (s). The origin is marked with 0. A horizontal green line is drawn at a negative value on the a-axis, representing constant negative acceleration (deceleration).	Object is under deceleration (negative value).

Type of Motion	Displacement	Velocity	Acceleration
Uniform Motion			
Non Uniform Motion			
Non Uniform Motion			

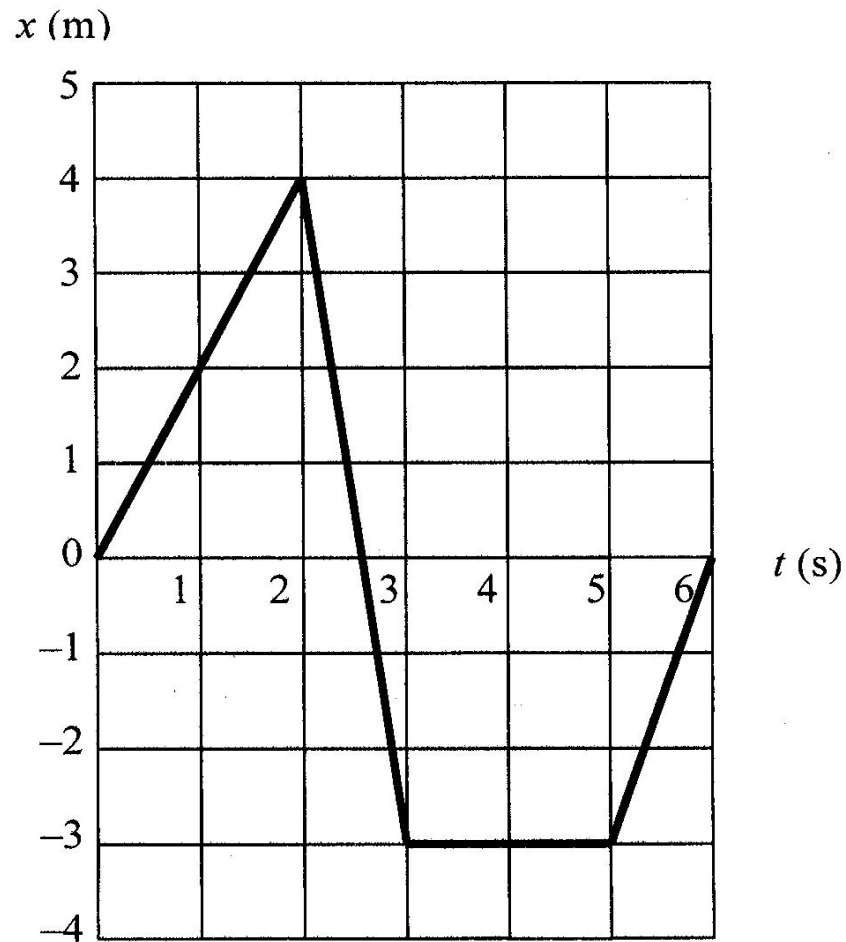
Example 7**FIGURE 5**

Figure 5 shows a graph of displacement x against time t of an object moving along x -axis. Calculate

- (a) Average velocity for the time interval, 1 s to 4 s.**
- (b) Average speed for the time interval, 1 s to 4 s.**
- (c) Instantaneous velocity at $t = 2.5$ s.**
- (d) Instantaneous acceleration at $t = 5.5$ s.**

Solution

$$\text{(a)} \quad \vec{v}_{\text{average}} = \frac{\Delta s}{\Delta t} = \frac{(-3) - 2}{4 - 1} = -1.67 \text{ m s}^{-1}$$

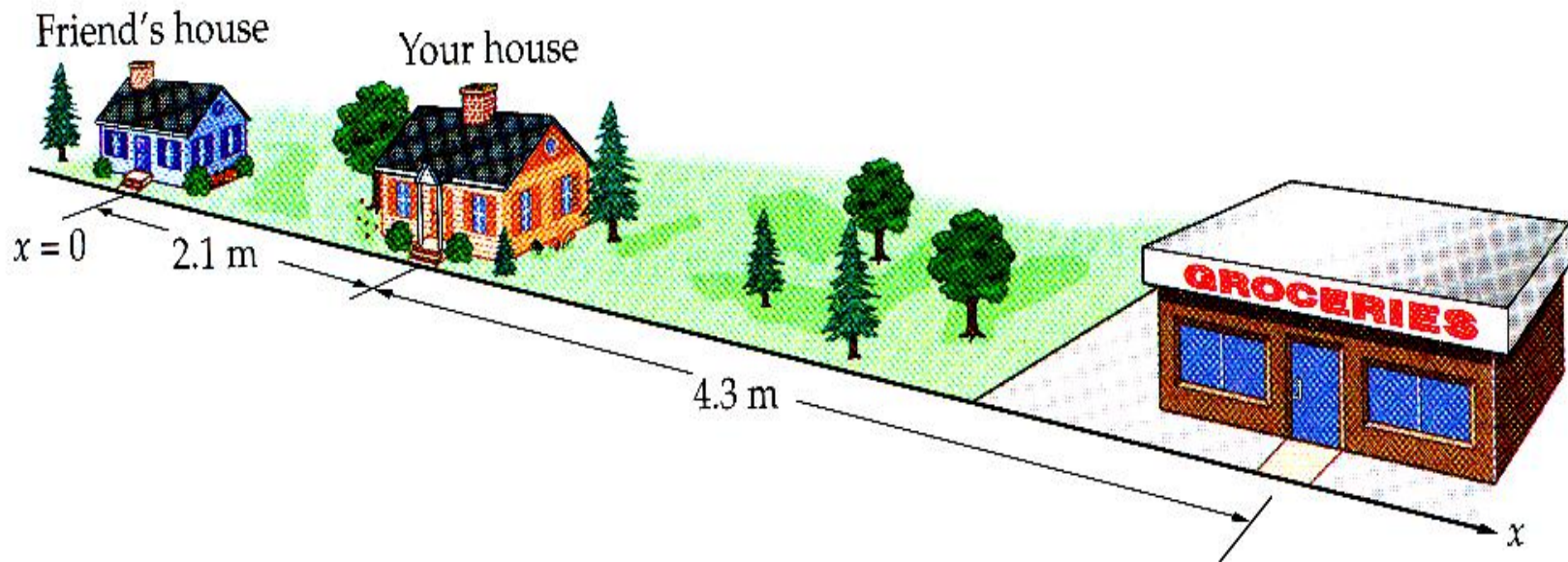
$$\text{(b)} \quad v_{\text{average}} = \frac{\Sigma x}{\Delta t} = \frac{2 + 4 + 3}{4 - 1} = 3 \text{ m s}^{-1}$$

$$\begin{aligned} \text{(c)} \quad v \Big|_{t=2.5\text{ s}} &= \frac{dx}{dt} = \text{gradient at } t = 2.5 \text{ s} \\ &= \frac{(-3) - 4}{3 - 2} \\ &= -7 \text{ m s}^{-1} \end{aligned}$$

$$\text{(d)} \quad \vec{a} = \frac{dv}{dt} = 0 \text{ m s}^{-2}$$

Follow up exercise

1. You walk from your house to friend's house then to the grocery shop. Calculate:
- (i) the distance traveled. (8.5m)
 - (ii) the displacement. (4.3m)



Follow up exercise

2. **An athlete runs four laps of a 400 m track.
What is the athlete's total displacement? (0m)**
3. **What is meant by distance and displacement?**

Solution

(1)(i) The distance travelled

$$= 2.1 + 2.1 + 4.3$$

$$= 8.5 \text{ m}$$

(ii) The displacement

$$= 2.1 + (-2.1) + 4.3$$

$$= 4.3 \text{ m}$$

(2) The total displacement = 0m

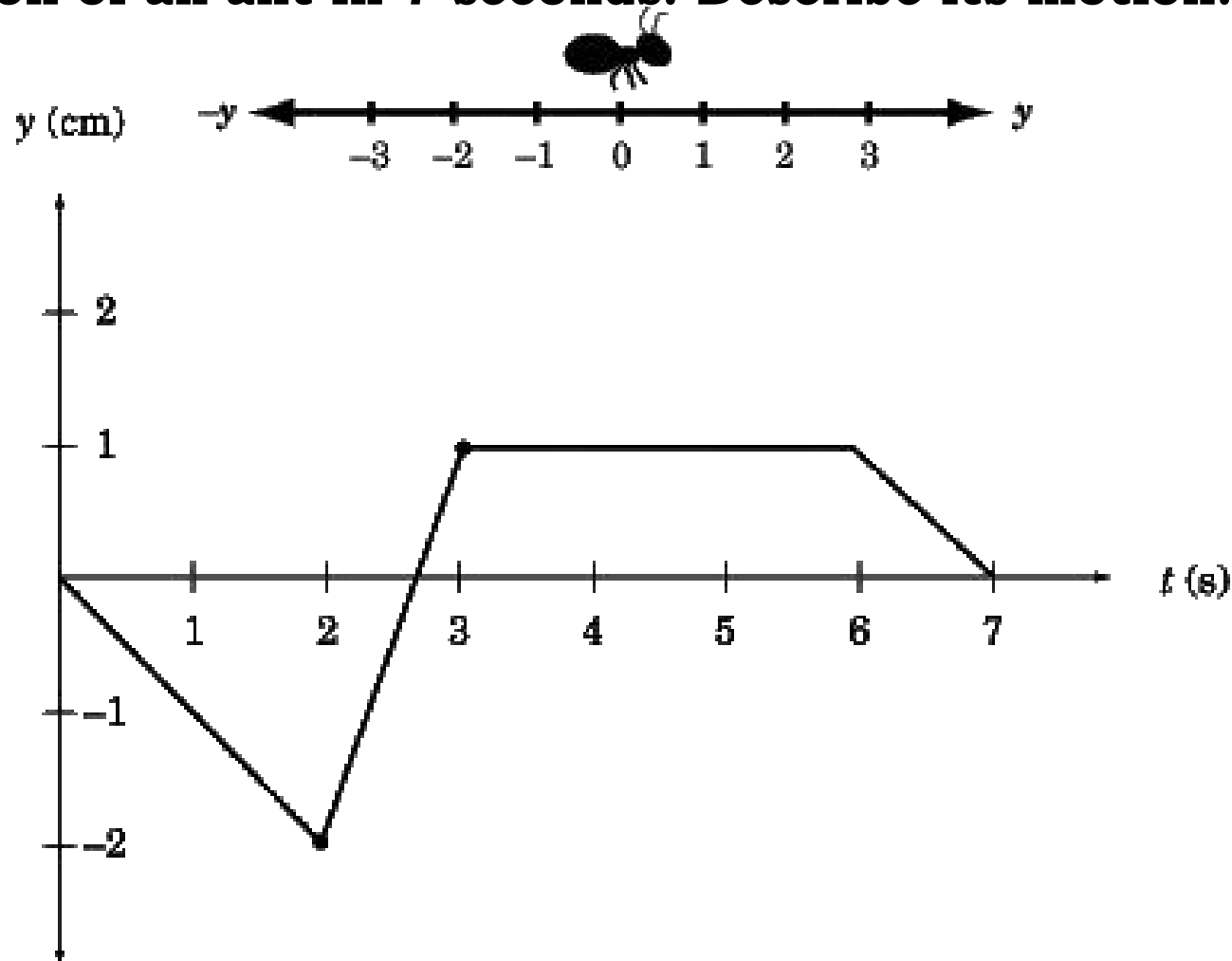
(3) Distance the **total length of travel** in moving from one location to another.

Displacement **straight line distance** (shortest distance) from the initial position to the final position of an object.

Follow up exercise

- 4. Can you accelerate a body without speeding up or slowing down? Is it possible?**
- 5. A car is traveling at 30 km h^{-1} to the north. Then it turns to the west without changing its speed. Is the car accelerating?**
- 6. How would you draw a displacement time graph for a stationary object?**
- 7. What would the gradient of a distance time graph represent?**
- 8. What does the area a speed-time graph represent?**

9. The distance – time graph in figure below represents the motion of an ant in 7 seconds. Describe its motion.



- 10. A body moves along the x -axis. Assume that a positive sign represents a direction to the right. The velocity, v of the body is related to time, t through the equation**

$$v = 2 - 3t^2$$

where v and t are measured in m s^{-1} and s respectively. $t = 0$ when $x = 0$.

Determine

(a) the displacement

(b) the acceleration

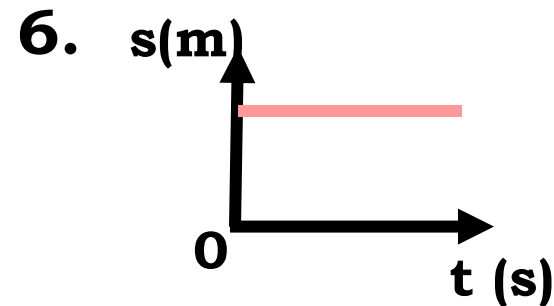
at the instant of time $t = 1 \text{ s}$

- 11. Define average speed and average velocity.**

Answer

4. **Yes, it is possible.**

5. **Yes, the car is accelerating.**



7. **Speed**

8. **Distance**

Answer

- 9.i. At $t=0\text{s}$ to $t=2\text{s}$ the ant move with constant speed 1 cms^{-1} to the back.**
- ii. At $t=2\text{s}$ to $t=3\text{s}$, the ant move forward with speed 3 cms^{-1} .**
- iii. At $t=3\text{s}$ to $t=6\text{s}$, the ant not moving.**
- iv. At $t=6\text{s}$ to $t=7\text{s}$, the move to the origin with speed 1 cms^{-1} .**

Answer

10. a. **displacement;**

$$v = \frac{dx}{dt}$$

$$x = \int v dt$$

$$= \int 2 - 3t^2 dt$$

$$= 2t - t^3$$

$$= 2(1) - 1^3$$

$$= 1 \text{ m}$$

Answer

10. b. acceleration;

$$a = \frac{dv}{dt}$$

$$= \frac{d}{dt}(2 - 3t^2)$$

$$= -3t(2)$$

$$= -6 \text{ ms}^{-2}$$

Answer**11. Average Speed, ($\bar{v}$)**

total distance traveled divided by the total time elapsed in traveling that distance.

$$\text{Average speed, } \bar{v} = \frac{\Sigma d}{\Sigma t}$$

Average velocity, $\vec{v}_{av}$

$$\vec{v}_{av} = \frac{\text{change in displacement}}{\text{time taken for the change}}$$

$$\vec{v}_{av} = \frac{\Delta s}{\Delta t} = \frac{s_2 - s_1}{t_2 - t_1}$$



2.2 UNIFORMLY ACCELERATED MOTION

At the end of this chapter, students should be able to:

(a) Apply equations of motion with uniform acceleration:

$$\begin{aligned}v &= u + at \\s &= ut + \frac{1}{2}at^2 \\v^2 &= u^2 + 2as\end{aligned}$$

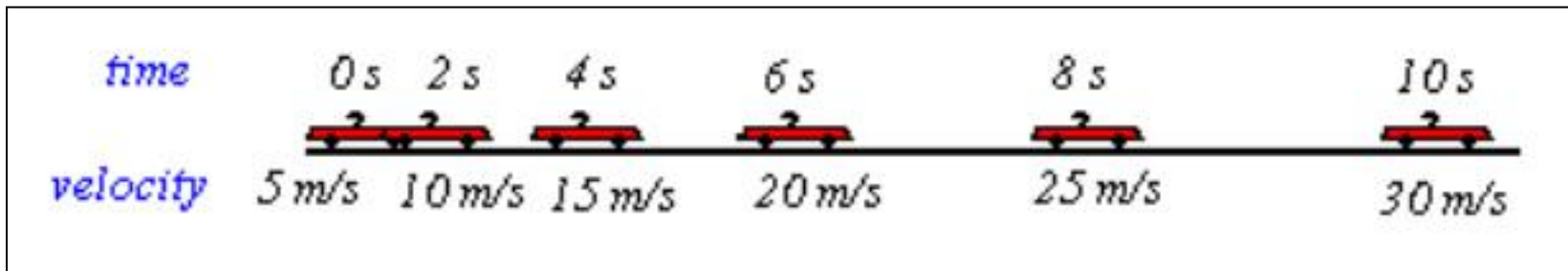
(b) Apply equations of motion for free fall:

$$\begin{aligned}v &= u - gt \\s &= ut - \frac{1}{2}gt^2 \\v^2 &= u^2 - 2gs\end{aligned}$$

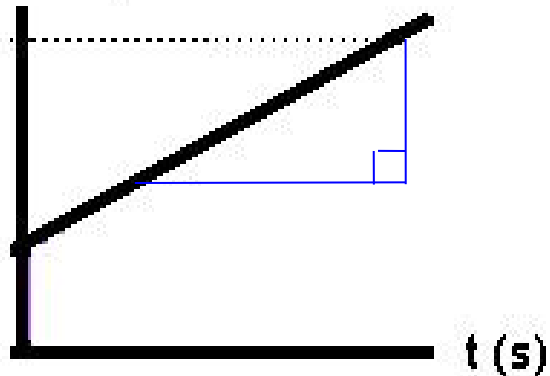
Uniform acceleration means the acceleration **does not depend on time** or always **constant**.

$$a = \frac{dv}{dt} = \text{constant}$$

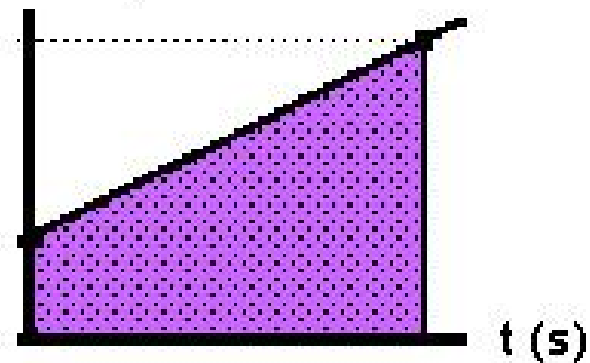
velocity changes at a uniform rate.



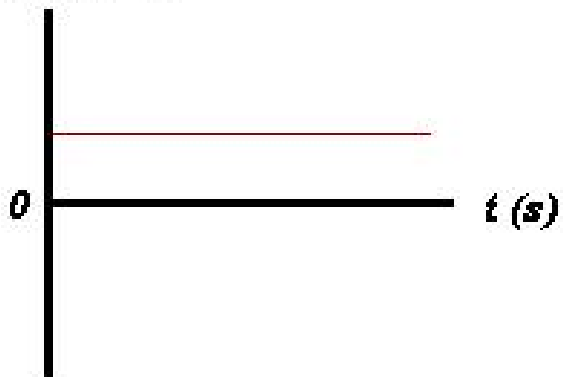
A $v - t$ plot is a straight line whose **gradient** is equal to **acceleration**.

$v \text{ (m s}^{-1}\text{)}$ 

**Acceleration =
gradient of the slope**

 $v \text{ (m s}^{-1}\text{)}$ 

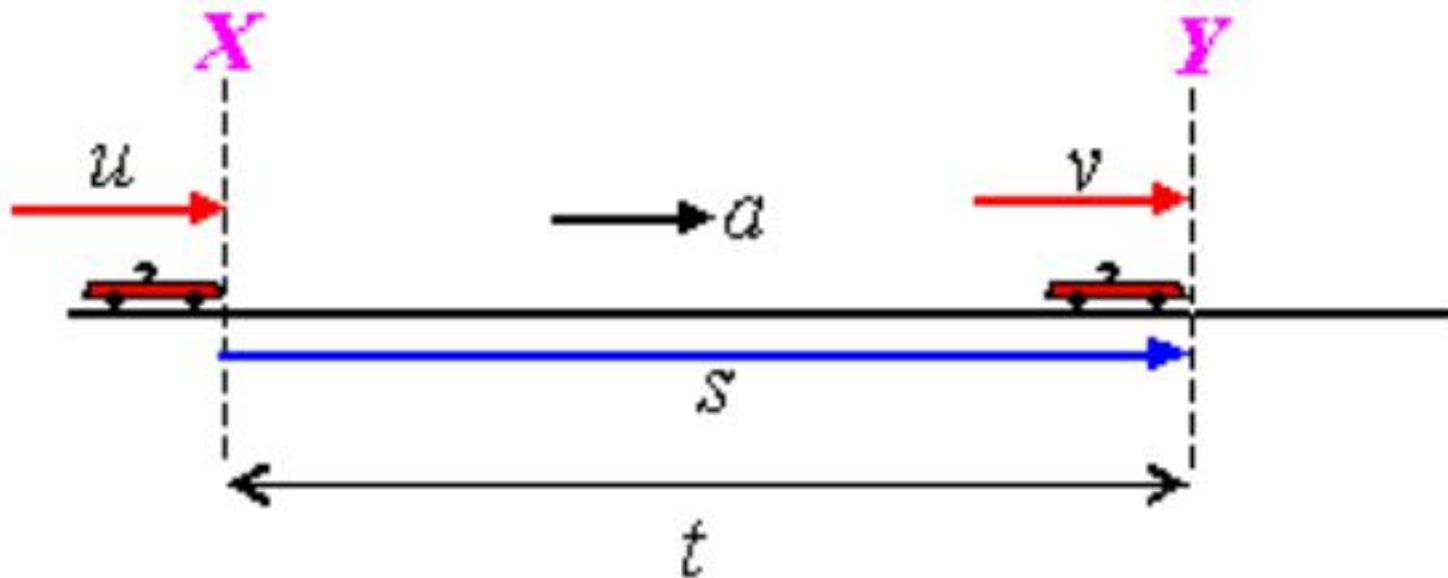
**Distance travelled
= area under the slope**

 $a \text{ (ms}^{-2}\text{)}$ 

**Acceleration = constant
value**

Kinematics Equation for uniform acceleration

Assume a car has **uniform acceleration** & consider the motion between X and Y :



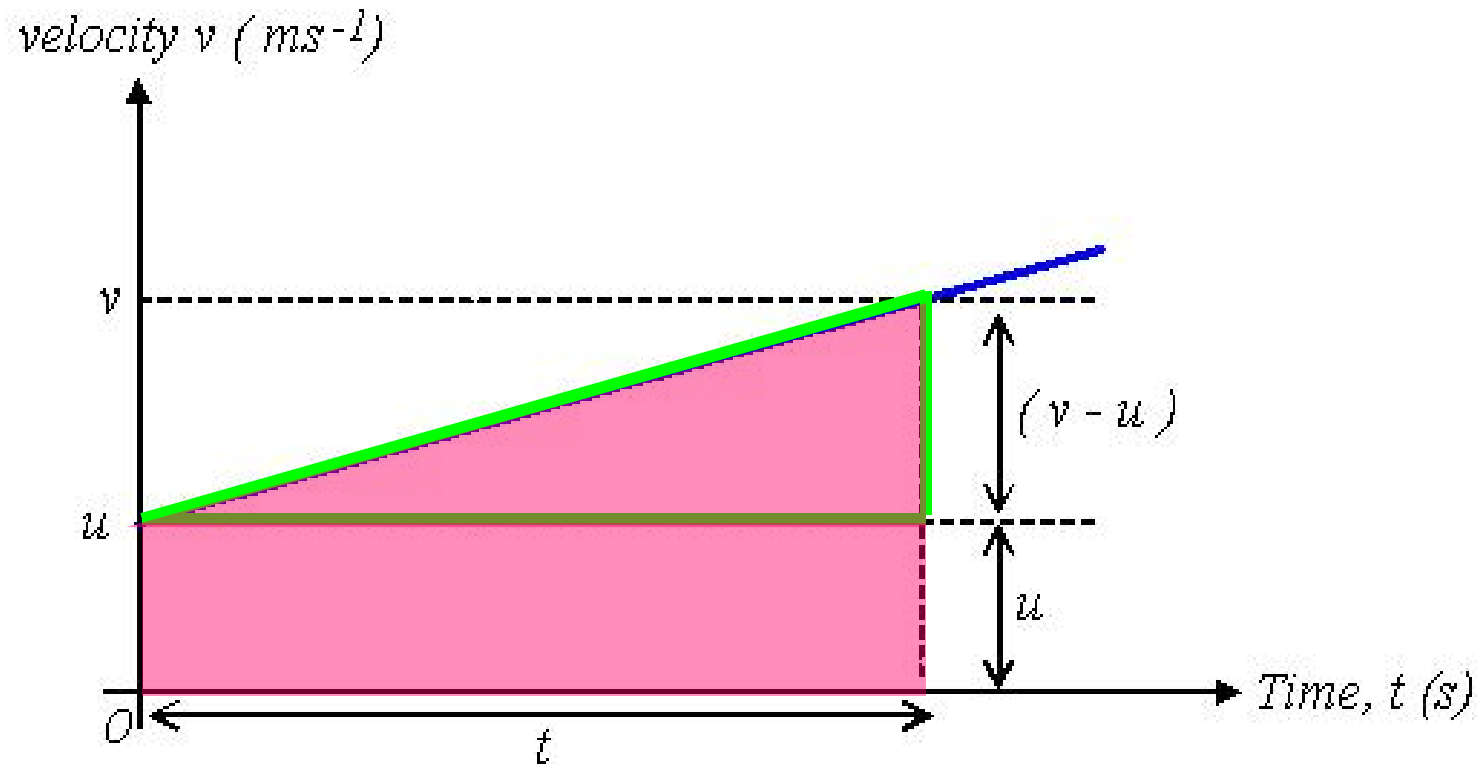
u = initial velocity (velocity on passing X)

v = final velocity (velocity on passing Y)

a = acceleration

s = displacement (in moving from X to Y)

t = time taken (to move from X to Y)

velocity – time graph for the car

Acceleration, a = gradient of graph $v - t$

$$a = \frac{v_2 - v_1}{t_2 - t_1} = \frac{v - u}{t}$$

rearranged to give:

$$v = u + a t \quad \dots (1)$$

Distance traveled, s = area under the graph

= area of trapezium

$$s = \frac{1}{2} (u + v) t \quad \dots (2)$$

Substitute (1) into (2) :

$$s = \frac{1}{2} (u + u + a t) t$$

$$s = u t + \frac{1}{2} a t^2 \quad \dots (3)$$

From (1) : $v = u + at$, get an expression for t :

$$t = \left(\frac{v - u}{a} \right)$$

Substitute into (2) :

$$s = \frac{1}{2} (v + u) \left(\frac{v - u}{a} \right)$$

$$= \frac{(v + u)(v - u)}{2a}$$

$$2as = v^2 - u^2$$

$$v^2 = u^2 + 2as \quad \dots \textbf{(4)}$$

Equations of motion

- Equations of motion with uniform acceleration :

$$s = \frac{1}{2}(v + u)t$$

$$s = ut + \frac{1}{2}at^2$$

$$v = u + at$$

$$v^2 = u^2 + 2as$$

Sign Convention

Quantity	To the left	To the right	Upward	Downward
Displacement, s	-	+	+	-
Velocity, v	-	+	+	-

Note:

- Acceleration : +ve, Deceleration: -ve
- Special case : uniform velocity,

$$u = v \Rightarrow a = 0, s = ut$$

Example 8

The driver of a pickup truck going 100 km h^{-1} applies the brakes, giving the truck a uniform deceleration of 6.50 m s^{-2} while it travels 20.0 m .

(a) What is the speed of the truck in kilometers per hour at the end of this distance ?

(b) How much time has elapsed ?

Solution

Given :

$$a = -6.5 \text{ m s}^{-2} \text{ (deceleration)}$$

$$s = 20 \text{ m}$$

$$u = 100 \text{ km h}^{-1}$$

$$= \frac{100(1000)\text{m}}{60(60)\text{s}} = 27.78 \text{ m s}^{-1}$$

(a) Final velocity , $v = ?$

$$v^2 = u^2 + 2as$$

$$v^2 = (27.78)^2 + 2(-6.5)(20)$$

$$v^2 = 511.7284$$

$$v = 22.62 \text{ m s}^{-1}$$

Convert to km h^{-1}

$$v = \frac{\frac{22.62}{1000} \text{ km}}{\frac{1}{60(60)} \text{ hour}}$$

$$v = 81.43 \text{ km h}^{-1}$$



(b) Assume : time elapsed, t

$$v = u + at$$

$$22.62 = 27.78 + (-6.5)t$$

$$-6.5t = -5.16$$

$$t = \frac{5.16}{6.5}$$

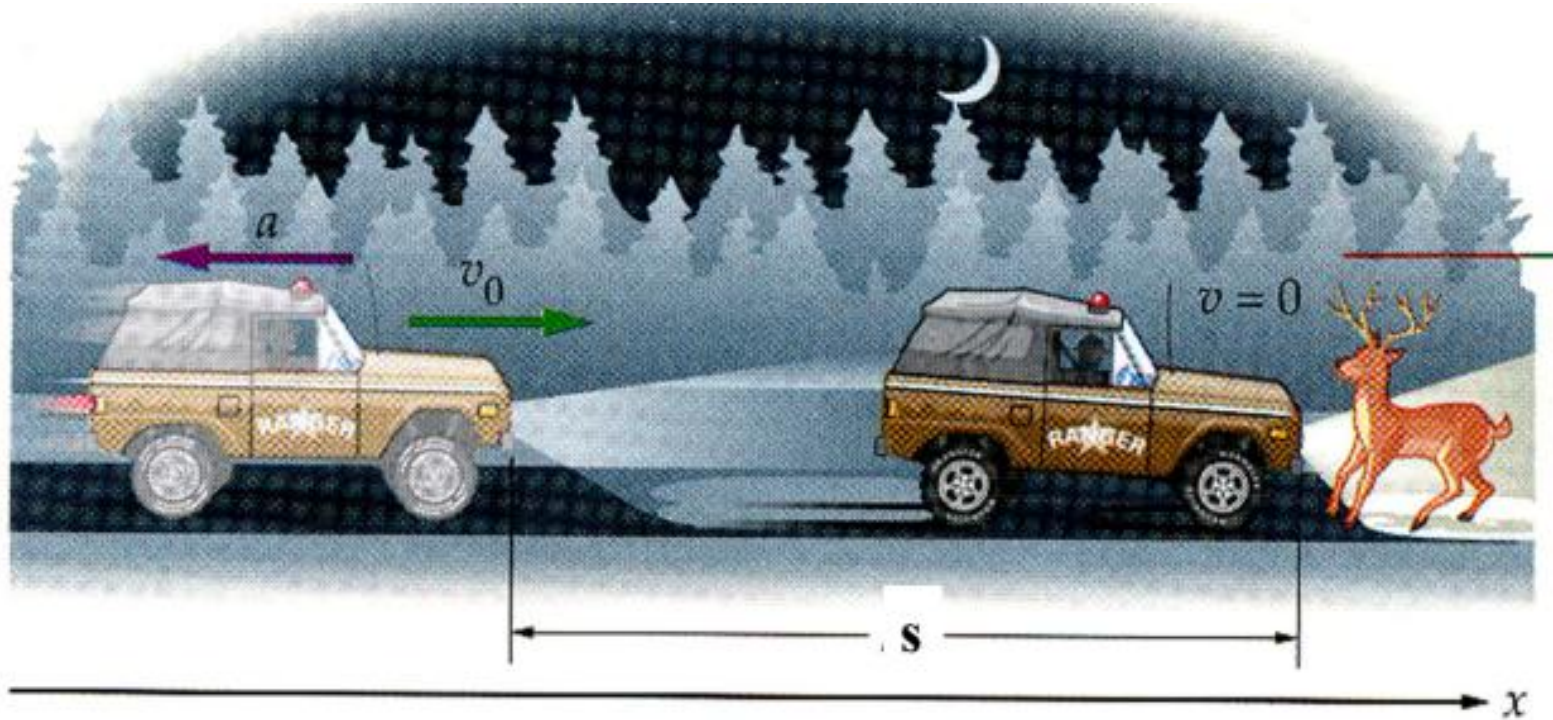
$$t = 0.794 \text{ s}$$



Example 9

A park ranger driving on a back country road suddenly sees a deer ‘frozen’ in his headlights. The ranger, who is driving at 11.4 m s^{-1} immediately applies the brakes and slows with an acceleration of 3.8 m s^{-2} .

- a) If the deer is 20.0 m from the ranger’s vehicle when the brakes are applied, how close does the ranger come to hitting the deer?**
- b) How much time is needed for the ranger’s vehicle to stop?**

Solution

Given: $u = 11.4 \text{ m s}^{-1}$; $a = -3.80 \text{ m s}^{-2}$

1st find the distance traveled before stopped

From: $v^2 = u^2 + 2as_2$

$$(0)^2 = (11.14)^2 + 2(-3.8)(s)$$

$$s = -\frac{129.96}{-7.6} = 17.1 \text{ m}$$

The distance between the stopped vehicle & deer:

$$x = 20 - 17.1 = 2.9 \text{ m}$$

(b) Time needed to stop = ?

From: $v = u + at$

$$0 = 11.4 + (-3.8)t$$

$$t = 3.0 \text{ s}$$

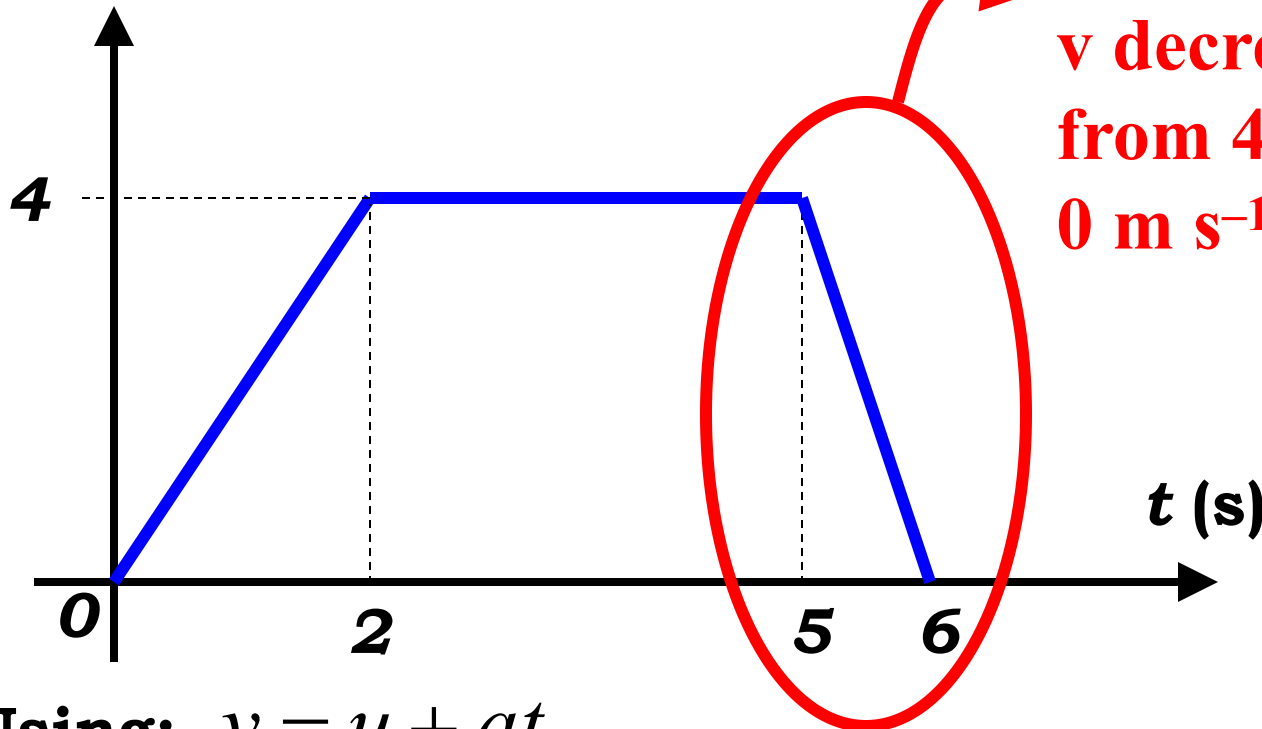
Example 10

A toy car moves with an acceleration of **2 m s^{-2} from rest for 2.0 s**. It then moves with **constant velocity for another 3.0 s**. It finally **comes to rest after another 1.0 s**.

- (a) Sketch a velocity-time graph to shown the motion of the toy car.
- (b) What is the velocity of the toy car after first 2 seconds ?
- (c) Calculate the deceleration of the car.
- (d) What is the total displacement of the car for the whole journey.
- (e) Sketch the acceleration-time graph for the motion of toy car.

Solution

(a)

 $v \text{ (m s}^{-1}\text{)}$ 

(b) Using: $v = u + at$

$$= 0 + 2(2)$$
$$= 4 \text{ m s}^{-1}$$



(c) Using: $v = u + at$

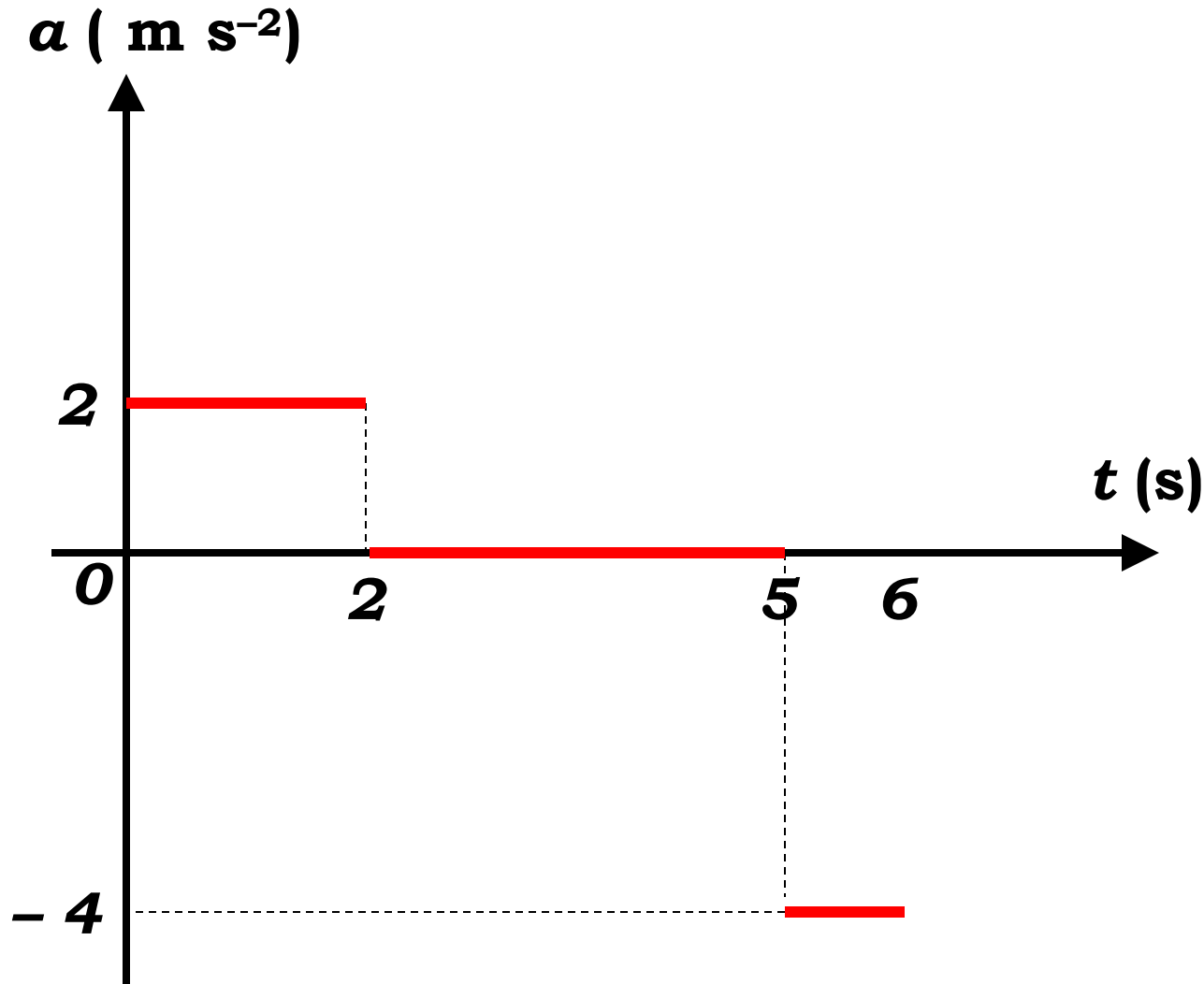
$$a = \frac{v - u}{t} = \frac{0 - 4}{1}$$

$$a = -4 \text{ m s}^{-2}$$

(d) Displacement, s = area under the graph

$$s = \frac{1}{2} (3 + 6)(4)$$

$$s = 18 \text{ m}$$

(e) Acceleration – time graph

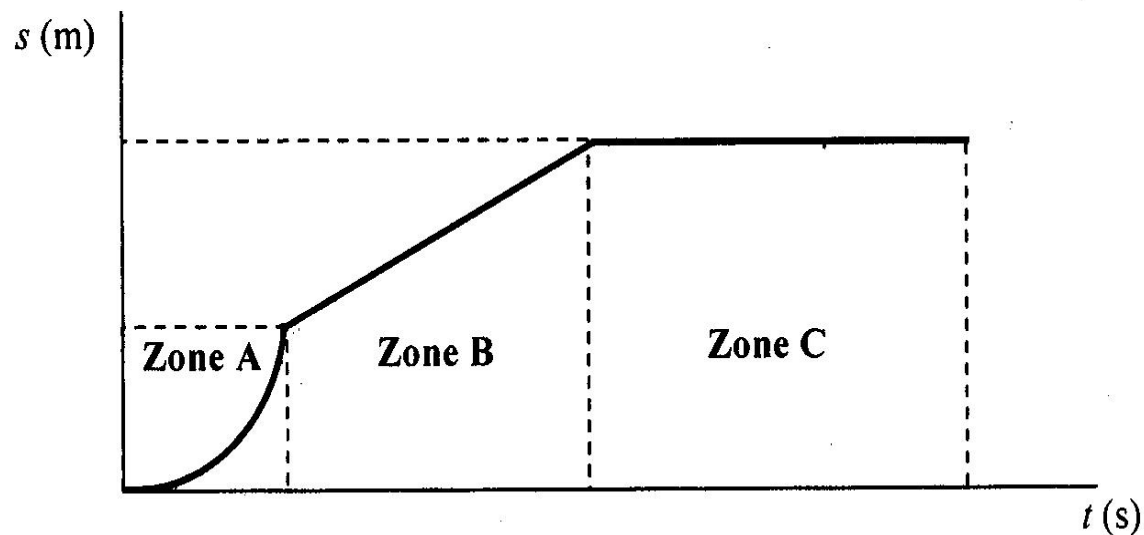
Follow up exercise

- 1** The speed limit in a school zone is 40 km h^{-1} . A driver traveling at this speed sees a child run onto the road 13 m ahead of his car. He applies the brakes and the car decelerates at a uniform rate of 8.0 m s^{-2} . If the driver's reaction time is 0.25 s , will the car stop before hitting the child ?
- 2**
 - (a) Is it possible for an object moving at non zero velocity has a zero acceleration? Explain.
 - (b) A car is capable of accelerating at 0.60 m s^{-2} . Calculate the time needed for this car to go from a speed of 5.5 m s^{-1} to a speed of 8.0 m s^{-1} .

3

Figure 2 shows a displacement – time graph of a car moving along a straight road.

Copy and complete Table 1 by stating any change (increase / decrease / constant / zero / no change) in the distance, speed and acceleration of the car for each zone.



Zone	Distance	Speed	Acceleration
A			
B			
C			

4 Show that $v = u + at$

5 Show that $s = ut + \frac{1}{2}at^2$

6 Show that $v^2 = u^2 + 2as$

7 An object moves along the x-axis. When it is at the centre of coordinate, its velocity is -6.0 m s^{-1} and its acceleration is 8.0 m s^{-2} .

Determine

(a) its position at $t = 2.0 \text{ s}$

(b) Its velocity at $t = 3.0 \text{ s}$

8 An object moves along a straight line with constant acceleration. Its initial velocity is 20 m s^{-1} . After 5.0 s , the velocity becomes 40 m s^{-1} . Determine the distance travelled during the third second.

Answer

- 1 Distance travel, in the reaction time interval;**

$$s = ut + \frac{1}{2}at^2$$

$$s = ut + \frac{1}{2}at^2$$

$$s = 11.11 (0.25) = 2.78 \text{ m}$$

Distance travel, during break being applied;

$$v^2 = u^2 + 2as$$

$$0^2 = 11.11^2 + 2(-8)s$$

$$s = 7.714 \text{ m}$$

$$\begin{aligned} \text{Total distance travel, } s &= 2.78 + 7.714 \\ &= 10.494 \text{ m} \end{aligned}$$

So, the car stop before hitting the child.

Answer**2**

a. Yes, when the object moving with constant velocity.

**b. $v = u + at$
 $8 = 5.5 + (0.6)t$
 $t = 4.17s$**

3

Zone	Distance	Speed	Acceleration
A	increase	increase	constant
B	increase	constant	zero
C	No change	zero	zero

Answer**7**

a. $s = ut + \frac{1}{2}at^2$

$$s = (-6)(2) + \frac{1}{2}(8)(2)^2$$
$$= 4 \text{ m}$$

b. $v = u + at$

$$v = (-6) + (8)(3)$$

$$v = 18 \text{ m s}^{-1}$$

Answer**8**

$$\mathbf{v = u + at}$$

$$\mathbf{40 = 20 + a(5)}$$

$$\mathbf{a = 4 \text{ m s}^{-2}}$$

$$s = ut + \frac{1}{2}at^2$$

$$s = (20)(3) + \frac{1}{2}(4)(3)^2$$

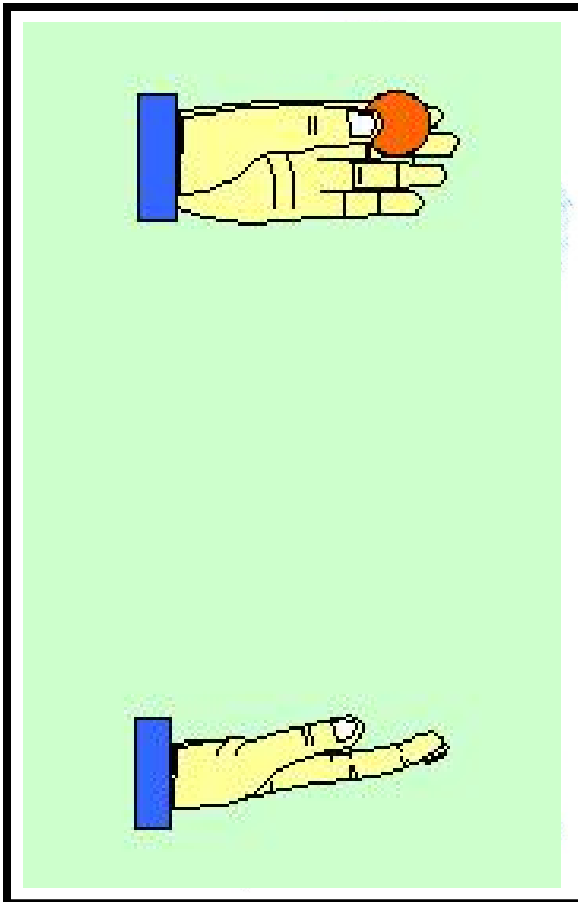
$$\mathbf{= 78 \text{ m}}$$

2.2(b) Free Falling Bodies

At the end of this chapter, students should be able to:

- a) Describe free falling body**
- b) Solve problems on free falling body**

❑ Free fall motion is linear vertical motion under the sole influence of gravity.



❑ The only force acting on the object is the **pull of gravity**.

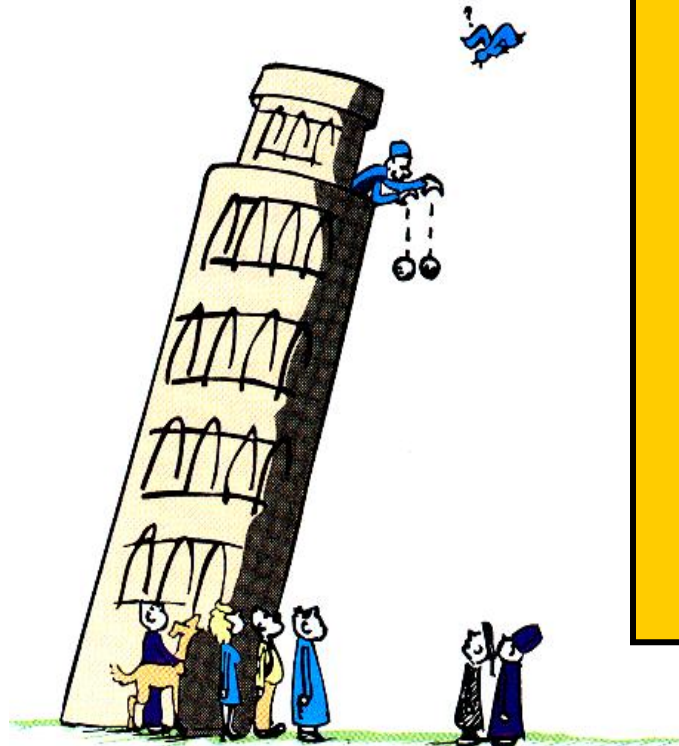
❑ Assumption : free falling objects do not encounter air resistance

❑ All free falling objects (on Earth) always accelerate downwards with an acceleration **$a = -g$**

❑ Value of **$g \approx 9.81 \text{ m s}^{-2}$** .

□ **g (vector quantity)** is given a **minus (-) sign** indicating that it is **always directed downward**.

□ Replace a with $(-g)$ into kinematics equation:



Galileo's Famous demonstration

Free Fall motion Equation

$$v = u - g t$$

$$v^2 = u^2 - 2 g s$$

$$s = u t - \frac{1}{2} g t^2$$

$$s = \frac{1}{2} [u + v] t$$

where

s : vertical displacement

u : initial velocity

v : final velocity

t : time interval

g : 9.81 m s^{-2}

□ Value of s , u , v may be (+) or (-) depending on the direction of motion.

Sign Convention

Quantity	Upward motion	Downward Motion
Displacement, s	+	-
Velocity, v	+	-
Acceleration due to gravity, g	-	-

At highest point, velocity, $v = 0 \text{ m s}^{-1}$;
acceleration, $a = -g$

Upwards

Journey:

displacement : +
velocity : +
acceleration : $-g$

Downwards

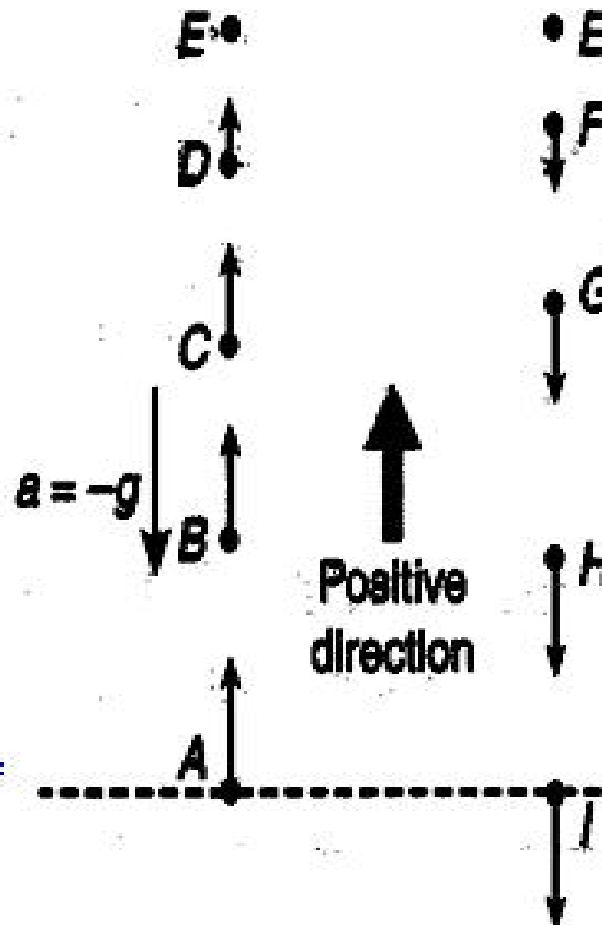
Journey:

Above release
point:

displacement : +
velocity : -
acceleration : $-g$

(Reference
level/origin)

release
point



Positive
direction

$s = 0$

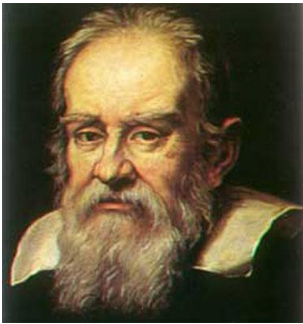
Below release point:

Displacement : -

velocity : -

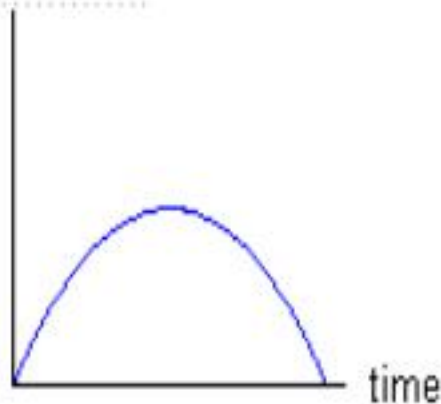
Acceleration : $-g$

Graphs of free fall object's position, velocity & acceleration as functions of time.

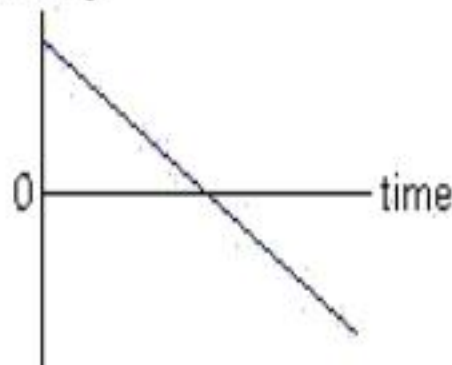


□ Motion graphs for an object thrown vertically upwards and then falling back to the ground.

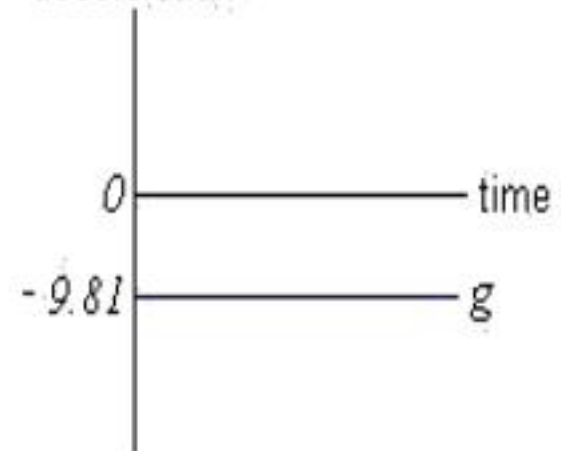
displacement



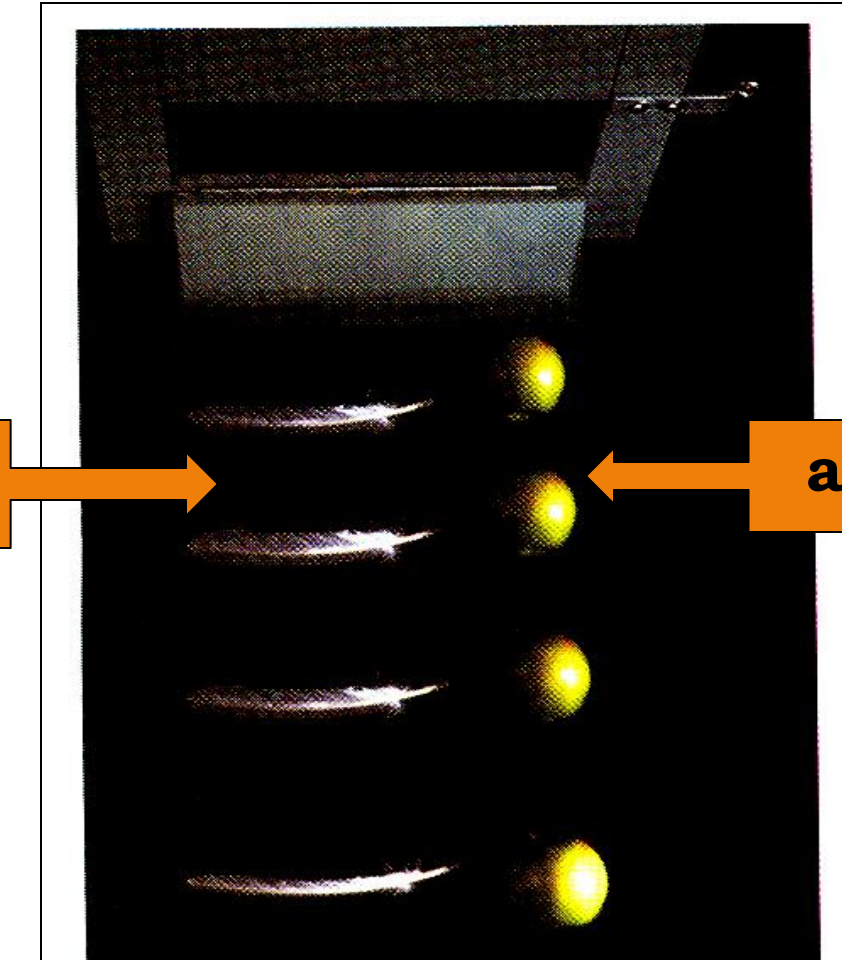
velocity



acceleration



a Feather



a Ball

In the absence of air resistance, all bodies fall with the same acceleration, regardless of their mass

Example 11

A student drops a ball from the top of a tall building, it takes 2.8 s for the ball to reach the ground.

- (a) What was the ball's speed just before hitting the ground ?
- (b) What is the height of the building ?

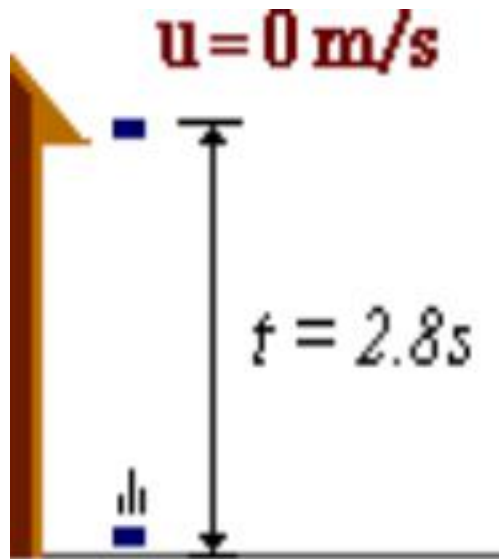
Solution

Given :

$$u = 0 \text{ m s}^{-1} \text{ (dropped) ;}$$

$$t = 2.8 \text{ s ;}$$

$$\text{free fall motion, } g = 9.81 \text{ ms}^{-2}$$



(a) $v = u - gt$

$$v = 0 - (9.81)(2.8)$$

$$v = 27.47 \text{ m s}^{-1}$$

*** (Minus sign – indicates that v is downward)**

(b) $s = ut - \frac{1}{2}gt^2$

$$s = 0 - \frac{1}{2}(9.81)(2.8)^2$$

$$s = -38.46 \text{ m}$$

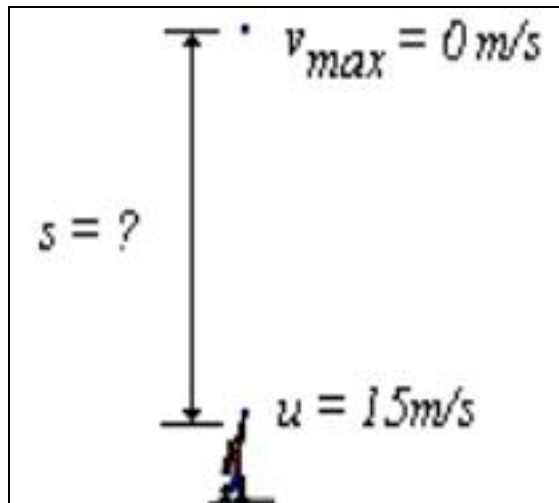
*** (Minus sign – indicates that displacement is downward)**

Example 12

A boy throws a stone straight upward with an initial speed of 15 m s^{-1} . What maximum height will the stone reach before falling back down ?

Solution

At maximum height, object's velocity is zero for an instant ($v = 0 \text{ m s}^{-1}$)



$$v^2 = u^2 - 2gs$$
$$(0)^2 = (15)^2 - 2(9.81)s$$
$$s = \frac{225}{19.62} = 11.47 \text{ m}$$

Example 13

A stone is **thrown vertically downward** at an initial speed of 14 ms^{-1} from a height of 65 m above the ground.

- (a) How far does the stone travel in 2 s ?
- (b) What is its velocity just before it hits the ground ?

Solution :

Given : $u = -14 \text{ m s}^{-1}$; $g = 9.81 \text{ m s}^{-2}$; $t = 2 \text{ s}$

(a) Using free fall equation :

$$s = ut - \frac{1}{2}gt^2$$

$$s = (-14)(2) - \frac{1}{2}(9.81)(2)^2$$

$$s = -28 - 19.62$$

$$s = -47.62 \text{ m}$$

(b) Assume the velocity just before hitting the ground = v

$$v^2 = u^2 - 2gs$$

$$v^2 = (-14)^2 - 2(9.81)(-65)$$

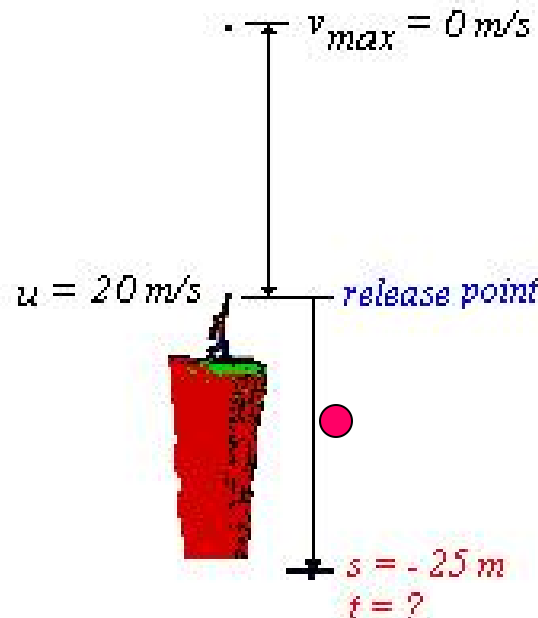
$$v^2 = 1471.3$$

$$v = -38.36 \text{ m s}^{-1}$$

Example 14

A small pebble is thrown upward from a cliff with an **initial velocity 20 m s^{-1}** . Calculate

- (a) **Maximum height** reached.
- (b) **Time** taken to reach a point 25 m below the initial point.

Solution

(a) At max height, $v = 0 \text{ m s}^{-1}$

From: $v^2 = u^2 - 2gs$

$$(0)^2 = (20)^2 - 2(9.81)s$$

$$s = \frac{400}{19.62}$$

$$s = 20.39 \text{ m (above release point)}$$

(b) Given : $s = -25 \text{ m}$ (below initial point)

From: $s = ut - \frac{1}{2}gt^2$

$$-25 = 20t - \frac{1}{2}(9.81)t^2$$

$$4.905t^2 - 20t - 25 = 0$$

From:

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-20) \pm \sqrt{(-20)^2 - 4(4.905)(-25)}}{2(4.905)}$$

$$= \frac{20 \pm 29.84}{9.81}$$

$$t = 5.1 \text{ s (+ve value only)}$$

Follow up exercise

1. An object is thrown vertically upwards from a point on the ground with speed u . Neglect air resistance. Determine
 - (a) the maximum height reached by the object
 - (b) the time taken to return to the starting point in terms of u and g
2. A ball is thrown vertically upwards from the top of a building at a speed of 15 m s^{-1} . If the height of the building is 200 m, determine
 - (a) the time taken by the ball to reach the ground
 - (b) the velocity when the ball reaches the ground

Follow up exercise

1. An object is thrown vertically upwards from a point on the ground with speed u . Neglect air resistance. Determine
- (a) the maximum height reached by the object
 - (b) the time taken to return to the starting point in terms of u and g

SOLUTION:

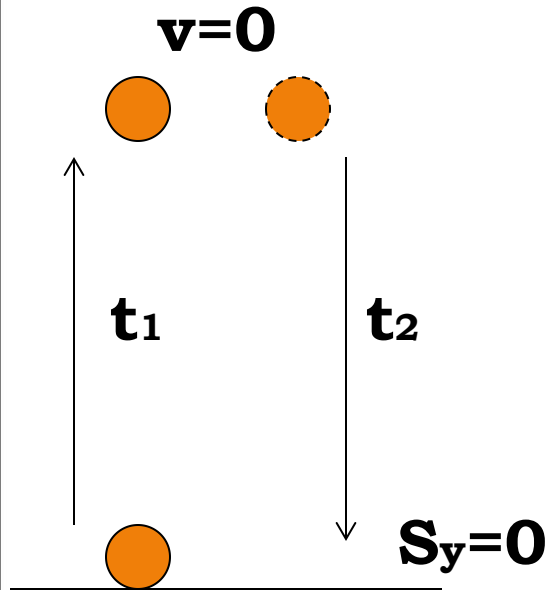
$$a) \ v^2 = u^2 - 2gS_y$$

at maximum height, $v = 0$

$$0 = u^2 - 2gS_y$$

$$2gS_y = u^2$$

$$S_y = \frac{u^2}{2g}$$

SOLUTION:**(b)**

$$v = u - gt$$

$$0 = u - gt$$

$$t = \frac{u}{g} = t_1 = t_2$$

time taken = total time

$$t_1 + t_2 = \left(\frac{u}{g} \right) + \left(\frac{u}{g} \right) = \frac{2u}{g}$$

OR

$$S_y = ut - \frac{1}{2}gt^2$$

$$0 = ut - \frac{1}{2}gt^2$$

$$ut = \frac{1}{2}gt^2$$

$$u = \frac{1}{2}gt$$

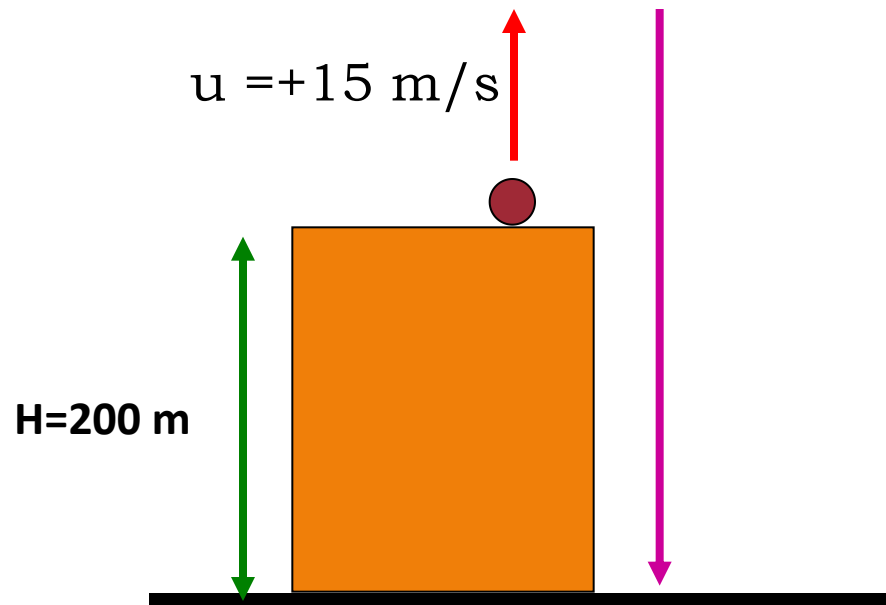
$$t = \frac{2u}{g}$$

Follow up exercise

2. A ball is thrown vertically upwards from the top of a building at a speed of 15 m s^{-1} . If the height of the building is 200 m, determine

- (a) the time taken by the ball to reach the ground
- (b) the velocity when the ball reaches the ground

SOLUTION:



SOLUTION:**Follow up exercise**

(a) Displacement, $S_y = -200 \text{ m}$, $u = 15 \text{ ms}^{-1}$,

Time of flight $= t$

$$\text{Using, } S_y = ut - \frac{1}{2}gt^2$$

$$-200 = (15)t - \frac{1}{2}(9.81)t^2$$

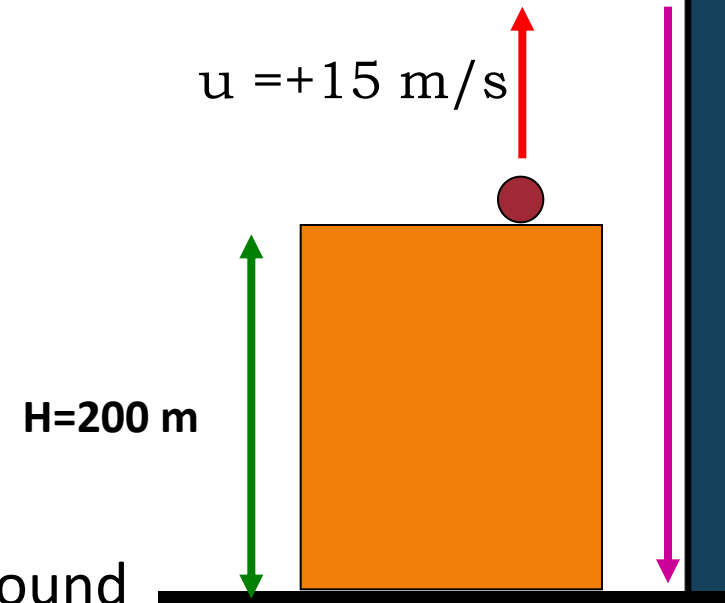
$$t = 8.1 \text{ s}$$

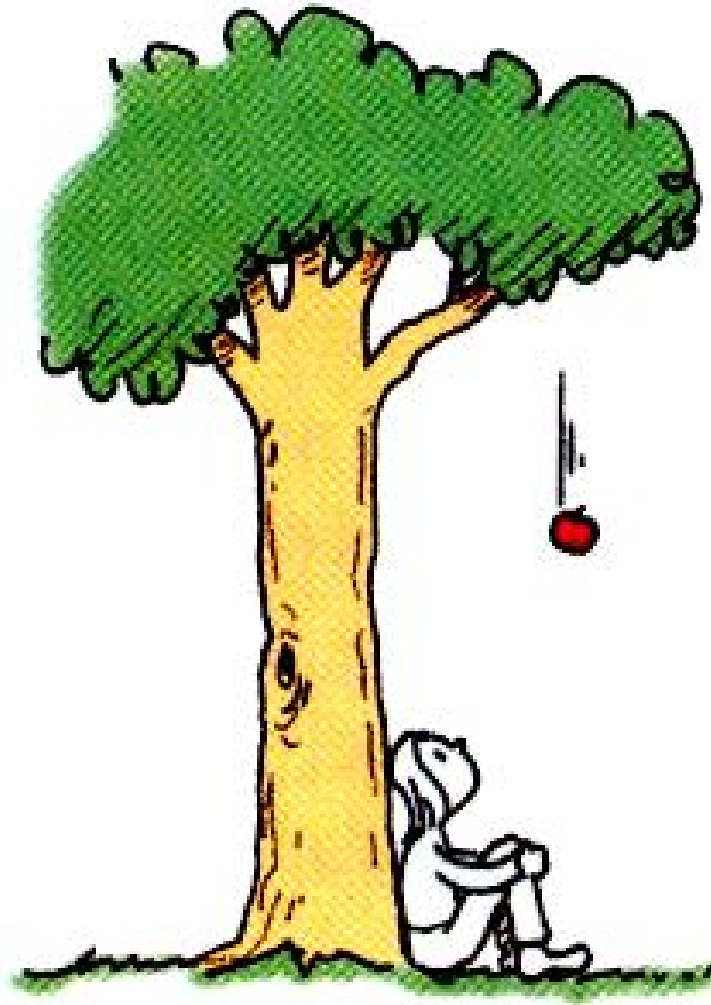
(b) velocity when the ball reaches the ground

$$v = u - gt$$

$$v = 15 - 9.81(8.1)$$

$$v = -65 \text{ ms}^{-1}$$





End of Chapter 2